

FPT UNIVERSITY - CAMPUS HO CHI MINH

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GROUP REPORT

Course: OSP201 - Open Source Platform and Network Administration

Class: IA1906

Lecturer: Mai Hoàng Đình

Service Monitoring System with Monit

Group 6

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Table of Contents

I. Introduction.....	2
II. Installation and Initial Configuration of Monit.....	3
1. Installation of Monit.....	3
a. Installation via YUM Package Manager.....	3
b. Installation from Source Code.....	4
2. Basic Configuration.....	8
a. Email Alert Configuration.....	8
b. Log and Web Interface Configuration.....	14
III. Advanced Monitoring Configuration.....	20
1. Process Monitoring.....	20
a. Syntax Specification.....	20
b. Example Configuration.....	21
2. File, Directory, and Filesystem Monitoring.....	26
a. Syntax Specification.....	26
b. Example Configuration.....	27
3. Program and Script Monitoring.....	33
a. Syntax Specification.....	33
b. Example Configuration.....	34
4. System Resource and Service Monitoring.....	36
a. CPU, Memory, and Load Average Monitoring.....	36
b. Network Interface Monitoring.....	38
c. Sendmail Service Monitoring.....	40
d. Apache Service Monitoring.....	42
IV. Extension: Centralized Monitoring with M/Monit.....	44
V. Conclusion.....	50
VI. References.....	51

I. Introduction

In Linux server environments, unexpected service failures can lead to downtime and serious disruptions. Manual monitoring is often inefficient, especially on systems that require continuous operation.

*To address this challenge, automated monitoring tools are necessary. Solutions like Zabbix or Prometheus can be complex to set up and resource-heavy. **Monit**, on the other hand, is a*

lightweight and easy-to-use tool designed for monitoring, error detection, and automatic recovery of services and system components.

Monit can monitor a wide range of system elements, including services (Apache, MySQL, Nginx), files and directories (file system changes), network connections (HTTP, SMTP), scripts and programs (similar to cron jobs), and system resources (CPU, memory, load average).

This project implements a monitoring system using Monit in a Linux environment. The main objectives are:

Continuously monitor the status of services and system resources.

Automatically detect failures and recover services.

Send real-time email alerts to system administrators.

This report provides step-by-step installation instructions, configuration examples, and use cases to demonstrate Monit's practicality and effectiveness.

This report is based on Monit version 5.2x.x (CentOS 7,8). Most configuration principles and operational behaviors remain consistent across newer versions (eg 5.35.x).

II. Installation and Initial Configuration of Monit

1. Installation of Monit

a. Installation via YUM Package Manager

(You need EPEL to install using this method)

sudo yum install monit

#Yum is a package manager for installing and updating software on CentOS.

```
File Edit View Search Terminal Help
[root@localhost ~]# vi /etc/yum.repos.d/epel.repo
[root@localhost ~]# sudo yum install monit
Loaded plugins: fastestmirror, langpacks
base                                     | 3.6 kB  00:00:00
epel                                     | 4.9 kB  00:00:00
extras                                  | 3.6 kB  00:00:00
updates                                | 3.6 kB  00:00:00
(1/2): epel/group_gz                   | 88 kB  00:00:01
(2/2): epel/primary_db                  | 6.7 MB  00:00:20
Loading mirror speeds from cached hostfile
Resolving Dependencies
--> Running transaction check
--> Package monit.x86_64 0:5.25.1-1.el7 will be installed
--> Finished Dependency Resolution

Dependencies Resolved

=====================================================================================================================================
Package                               Arch             Version           Repository        Size
=====================================================================================================================================
Installing:
monit                                 x86_64            5.25.1-1.el7      epel               318 k
Transaction Summary
-----
Install 1 Package

Total download size: 318 k
Installed size: 818 k
Is this ok [y/d/N]: y
Downloading packages:
warning: /var/cache/yum/x86_64/7/epel/packages/monit-5.25.1-1.el7.x86_64.rpm: Header V3 RSA/SHA256 Signature, key ID 352c64e5: NOKEY:01 ETA
Public key for monit-5.25.1-1.el7.x86_64.rpm is not installed
monit-5.25.1-1.el7.x86_64.rpm           | 318 kB  00:00:02
Retrieving key from file:///etc/pki/rpm-gpg/RPM-GPG-KEY-EPEL-7
Importing GPG key 0x352C64E5:
  Userid : "Fedora EPEL (7) <epel@fedoraproject.org>"
  Fingerprint: 91e9 7d7c 4a5e 96f1 7f3e 888f 6a2f aea2 352c 64e5
  Package : epel-release-7-11.noarch (installed)
  From    : /etc/pki/rpm-gpg/RPM-GPG-KEY-EPEL-7
Is this ok [y/N]: y
Running transaction check
Running transaction test
Transaction test succeeded
Running transaction

Package                               Arch             Version           Repository        Size
=====================================================================================================================================
Installing:
monit                                 x86_64            5.25.1-1.el7      epel               318 k
Transaction Summary
-----
Install 1 Package

Total download size: 318 k
Installed size: 818 k
Is this ok [y/d/N]: y
Downloading packages:
warning: /var/cache/yum/x86_64/7/epel/packages/monit-5.25.1-1.el7.x86_64.rpm: Header V3 RSA/SHA256 Signature, key ID 352c64e5: NOKEY:01 ETA
Public key for monit-5.25.1-1.el7.x86_64.rpm is not installed
monit-5.25.1-1.el7.x86_64.rpm           | 318 kB  00:00:02
Retrieving key from file:///etc/pki/rpm-gpg/RPM-GPG-KEY-EPEL-7
Importing GPG key 0x352C64E5:
  Userid : "Fedora EPEL (7) <epel@fedoraproject.org>"
  Fingerprint: 91e9 7d7c 4a5e 96f1 7f3e 888f 6a2f aea2 352c 64e5
  Package : epel-release-7-11.noarch (installed)
  From    : /etc/pki/rpm-gpg/RPM-GPG-KEY-EPEL-7
Is this ok [y/N]: y
Running transaction check
Running transaction test
Transaction test succeeded
Running transaction
Warning: RPMDB altered outside of yum.
Installing : monit-5.25.1-1.el7.x86_64
Verifying  : monit-5.25.1-1.el7.x86_64

Installed:
monit.x86_64 0:5.25.1-1.el7

Complete!
[root@localhost ~]# which monit
/usr/bin/monit
[root@localhost ~]# monit -V
This is Monit version 5.25.1
Built with ssl, with ipv6, with compression, with pam and with large files
Copyright (C) 2001-2017 Tildeslash Ltd. All Rights Reserved.
[root@localhost ~]#
```

```
File Edit View Search Terminal Help
Dependencies Resolved

=====================================================================================================================================
Package                               Arch             Version           Repository        Size
=====================================================================================================================================
Installing:
monit                                 x86_64            5.25.1-1.el7      epel               318 k
Transaction Summary
-----
Install 1 Package

Total download size: 318 k
Installed size: 818 k
Is this ok [y/d/N]: y
Downloading packages:
warning: /var/cache/yum/x86_64/7/epel/packages/monit-5.25.1-1.el7.x86_64.rpm: Header V3 RSA/SHA256 Signature, key ID 352c64e5: NOKEY:01 ETA
Public key for monit-5.25.1-1.el7.x86_64.rpm is not installed
monit-5.25.1-1.el7.x86_64.rpm           | 318 kB  00:00:02
Retrieving key from file:///etc/pki/rpm-gpg/RPM-GPG-KEY-EPEL-7
Importing GPG key 0x352C64E5:
  Userid : "Fedora EPEL (7) <epel@fedoraproject.org>"
  Fingerprint: 91e9 7d7c 4a5e 96f1 7f3e 888f 6a2f aea2 352c 64e5
  Package : epel-release-7-11.noarch (installed)
  From    : /etc/pki/rpm-gpg/RPM-GPG-KEY-EPEL-7
Is this ok [y/N]: y
Running transaction check
Running transaction test
Transaction test succeeded
Running transaction
Warning: RPMDB altered outside of yum.
Installing : monit-5.25.1-1.el7.x86_64
Verifying  : monit-5.25.1-1.el7.x86_64

Installed:
monit.x86_64 0:5.25.1-1.el7

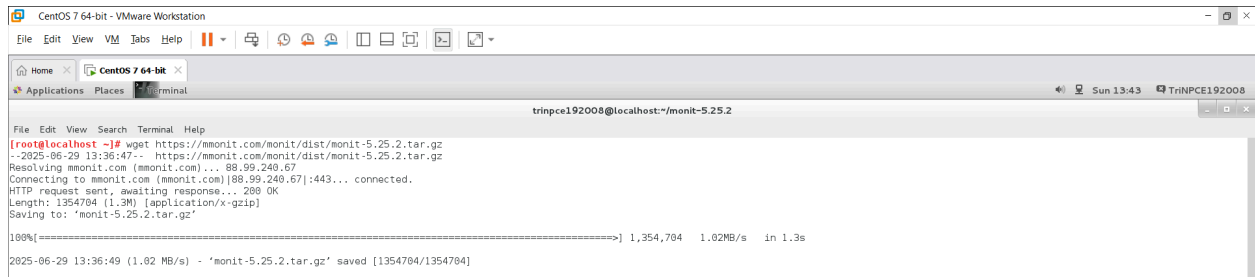
Complete!
[root@localhost ~]# which monit
/usr/bin/monit
[root@localhost ~]# monit -V
This is Monit version 5.25.1
Built with ssl, with ipv6, with compression, with pam and with large files
Copyright (C) 2001-2017 Tildeslash Ltd. All Rights Reserved.
[root@localhost ~]#
```

b. Installation from Source Code

Step 1:

wget https://mmonit.com/monit/dist/monit-5.25.2.tar.gz

#Downloads a file from the internet.



```
CentOS 7 64-bit - VMware Workstation
File Edit View VM Tabs Help
CentOS 7 64-bit
Applications Places terminal
trinpc192008@localhost:~/monit-5.25.2

File Edit View Search Terminal Help
[root@localhost ~]# wget https://mmonit.com/monit/dist/monit-5.25.2.tar.gz
--2025-06-29 13:36:47-- https://mmonit.com/monit/dist/monit-5.25.2.tar.gz
Resolving mmonit.com (mmonit.com)... 88.99.240.67
Connecting to mmonit.com (mmonit.com)|88.99.240.67|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1354704 (1.3M) [application/x-gzip]
Saving to: 'monit-5.25.2.tar.gz'

100%[=====] 1,354,704 1.02MB/s in 1.3s

2025-06-29 13:36:49 (1.02 MB/s) - 'monit-5.25.2.tar.gz' saved [1354704/1354704]
```

If error: SSL error due to missing CA certificate

Fix: **sudo yum install ca-certificates -y**

Install trusted certificate authorities (for secure HTTPS)

sudo update-ca-trust force-enable

Enable the updated certificate system

```
[root@localhost trinpc192008]# wget https://mmonit.com/monit/dist/monit-5.9.0.tar.gz
--2025-06-29 12:14:26-- https://mmonit.com/monit/dist/monit-5.9.0.tar.gz
Resolving mmonit.com (mmonit.com)... 88.99.240.67
Connecting to mmonit.com (mmonit.com)|88.99.240.67|:443... connected.
ERROR: cannot verify mmonit.com's certificate, issued by '/C=US/O=Network Solutions L.L.C./CN=Network Solutions RSA DV SSL CA 3':
Unable to locally verify the issuer's authority.
To connect to mmonit.com insecurely, use '--no-check-certificate'.
```

Step 2: Install monit from downloaded package

tar xzvf monit-5.25.2.tar.gz *#Extracts the downloaded Monit package.*

cd monit-5.25.2 *#Enters the extracted Monit folder.*

./configure *#checks system dependencies.*

make *#Compiles Monit from source code.*

sudo make install *#Installs Monit after compilation.*

which monit *#check installation location.*

monit -V *#Shows the installed Monit version.*

CentOS 7 64-bit - VMware Workstation

File Edit View VM Tabs Help

CentOS 7 64-bit

Applications Places Terminal

trinpce192008@localhost:~/monit-5.25.2

```
File Edit View Search Terminal Help
anaconda-ks.cfg monit-5.35.2 test-blogd.zip test.txt.gpg test-zip-2.txt
monit-5.25.2.tar.gz monit-5.35.2.tar.gz test.txt test-zip-1.txt test-zip-3.txt
[root@localhost ~]# tar xzvf monit-5.25.2.tar.gz
monit-5.25.2/
monit-5.25.2/libmonit/
monit-5.25.2/libmonit/configure.ac
monit-5.25.2/libmonit/Makefile.am
monit-5.25.2/libmonit/config/
monit-5.25.2/libmonit/config/compile
monit-5.25.2/libmonit/config/missing
monit-5.25.2/libmonit/config/config.sub
monit-5.25.2/libmonit/config/install-sh
monit-5.25.2/libmonit/config/config.guess
monit-5.25.2/libmonit/config/ltmain.sh
monit-5.25.2/libmonit/COPYING
monit-5.25.2/libmonit/Makefile.in
monit-5.25.2/libmonit/README
monit-5.25.2/libmonit/test/
monit-5.25.2/libmonit/test/OutputStreamTest.c
monit-5.25.2/libmonit/test/Makefile
monit-5.25.2/libmonit/test/InputStreamTest.c
monit-5.25.2/libmonit/test/Makefile.am
monit-5.25.2/libmonit/test/ExceptionTest.c
monit-5.25.2/libmonit/test/FileTest.c
monit-5.25.2/libmonit/test/ListTest.c
monit-5.25.2/libmonit/test/Makefile.in
monit-5.25.2/libmonit/test/TimeTest.c
monit-5.25.2/libmonit/test/LinkTest.c
monit-5.25.2/libmonit/test/DirTest.c
monit-5.25.2/libmonit/test/data/
monit-5.25.2/libmonit/test/data/stream.data
monit-5.25.2/libmonit/test/StrTest.c
monit-5.25.2/libmonit/test/SystemTest.c
monit-5.25.2/libmonit/test/CommandTest.c
monit-5.25.2/libmonit/test/NetTest.c
monit-5.25.2/libmonit/test/test.sh
monit-5.25.2/libmonit/test/StringBufferTest.c
monit-5.25.2/libmonit/bootstrap
monit-5.25.2/libmonit/src/
monit-5.25.2/libmonit/src/util/
monit-5.25.2/libmonit/src/util/StringBuffer.h
monit-5.25.2/libmonit/src/util/StringBuffer.c
monit-5.25.2/libmonit/src/util/Str.h
monit-5.25.2/libmonit/src/util/List.c
monit-5.25.2/libmonit/src/util/Str.c
```

Q Search

ENG 1:45:16 PM 6/29/2025

CentOS 7 64-bit - VMware Workstation

File Edit View VM Tabs Help

CentOS 7 64-bit

Applications Places Terminal

trinpce192008@localhost:~/monit-5.25.2

```
File Edit View Search Terminal Help
monit-5.25.2/m4/itsugar.m4
monit-5.25.2/configure
[root@localhost ~]# cd monit-5.25.2
[root@localhost monit-5.25.2]# ./configure
checking for a BSD-compatible install... /usr/bin/install -c
checking whether build environment is sane... yes
checking for a thread-safe mkdir -p... /usr/bin/mkdir -p
checking for gawk... gawk
checking whether make sets $(MAKE)... yes
checking whether make supports nested variables... yes
checking for gcc... gcc
checking whether the C compiler works... yes
checking for C compiler default output file name... a.out
checking for suffix of executables...
checking whether we are cross compiling... no
checking for suffix of object files... o
checking whether we are using the GNU C compiler... yes
checking whether gcc accepts -g... yes
checking for gcc option to accept ISO C89... none needed
checking whether gcc understands -c and -o together... yes
checking for style of include used by make... GNU
checking dependency style of gcc... none
checking whether make sets $(MAKE)... (cached) yes
checking for bison... no
checking for byacc... no
checking for yacc... no
checking for flex... no
checking for pod2man... /usr/bin/pod2man
checking build system type... x86_64-pc-linux-gnu
checking host system type... x86_64-pc-linux-gnu
checking how to print strings... printf
checking for a sed that does not truncate output... /usr/bin/sed
checking for grep that handles long lines and -e... /usr/bin/grep
checking for egrep... /usr/bin/grep -E
checking for fgrep... /usr/bin/grep -F
checking for ld used by gcc... /usr/bin/ld
checking if the linker (/usr/bin/ld) is GNU ld... yes
checking for BSD- or MS-compatible name lister (nm)... /usr/bin/nm -B
checking the name lister (/usr/bin/nm -B) interface... BSD nm
checking whether ln -s works... yes
checking the maximum length of command line arguments... 1572864
checking how to convert x86_64-pc-linux-gnu file names to x86_64-pc-linux-gnu format... func_convert_file_noop
checking how to convert x86_64-pc-linux-gnu file names to toolchain format... func_convert_file_noop
checking for /usr/bin/ld option to reload object files... -r
checking for objdump... objdump
```

Q Search

ENG 1:46:39 PM 6/29/2025

```
CentOS 7 64-bit - VMware Workstation
File Edit View VM Tabs Help
CentOS 7 64-bit
Applications Places Terminal
trinpc192008@localhost:~/monit-5.25.2

File Edit View Search Terminal Help
[root@localhost monit-5.25.2]# make
make all-recursive
make[1]: Entering directory '/root/monit-5.25.2'
Making all in libmonit
make[2]: Entering directory '/root/monit-5.25.2/libmonit'
Making all in .
make[3]: Entering directory '/root/monit-5.25.2/libmonit'
/bin/sh ./libtool --tagCC --mode=compile gcc -DHAVE_CONFIG_H -I. -I./src -I./src -I./src/exceptions -I./src/lo -I./src/net -I./src/st
statistics -I./src/util -I./src/thread -Wno-address -Wno-pointer-sign -g -O2 -D_REENTRANT -Wall -Wunused -Wno-unused-label -fsigned-char
-D_GNU_SOURCE -std=c99 -c -o src/Bootstrap.lo src/Bootstrap.c
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I./src -I./src -I./src/exceptions -I./src/lo -I./src/net -I./src/statistics -I./src/util -I./src
/thread -Wno-address -Wno-pointer-sign -g -O2 -D_REENTRANT -Wall -Wunused -Wno-unused-label -fsigned-char -D_GNU_SOURCE -std=c99 -c src/B
ootstrap.c -fPIC -DPIC -o src/.libs/Bootstrap.o
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I./src -I./src -I./src/exceptions -I./src/lo -I./src/net -I./src/statistics -I./src/util -I./src
/thread -Wno-address -Wno-pointer-sign -g -O2 -D_REENTRANT -Wall -Wunused -Wno-unused-label -fsigned-char -D_GNU_SOURCE -std=c99 -c src/B
ootstrap.c -o src/Bootstrap.o >/dev/null 2>&1
/bin/sh ./libtool --tagCC --mode=compile gcc -DHAVE_CONFIG_H -I. -I./src -I./src -I./src/exceptions -I./src/lo -I./src/net -I./src/st
statistics -I./src/util -I./src/thread -Wno-address -Wno-pointer-sign -g -O2 -D_REENTRANT -Wall -Wunused -Wno-unused-label -fsigned-char
-D_GNU_SOURCE -std=c99 -c -o src/exceptions/assert.lo src/exceptions/assert.c
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I./src -I./src -I./src/exceptions -I./src/lo -I./src/net -I./src/statistics -I./src/util -I./src
/thread -Wno-address -Wno-pointer-sign -g -O2 -D_REENTRANT -Wall -Wunused -Wno-unused-label -fsigned-char -D_GNU_SOURCE -std=c99 -c src/e
xceptions/assert.c -fPIC -DPIC -o src/exceptions/.libs/assert.o
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I./src -I./src -I./src/exceptions -I./src/lo -I./src/net -I./src/statistics -I./src/util -I./src
/thread -Wno-address -Wno-pointer-sign -g -O2 -D_REENTRANT -Wall -Wunused -Wno-unused-label -fsigned-char -D_GNU_SOURCE -std=c99 -c src/e
xceptions/assert.c -o src/exceptions/assert.o >/dev/null 2>&1
/bin/sh ./libtool --tagCC --mode=compile gcc -DHAVE_CONFIG_H -I. -I./src -I./src -I./src/exceptions -I./src/lo -I./src/net -I./src/st
statistics -I./src/util -I./src/thread -Wno-address -Wno-pointer-sign -g -O2 -D_REENTRANT -Wall -Wunused -Wno-unused-label -fsigned-char
-D_GNU_SOURCE -std=c99 -c -o src/exceptions/Exception.lo src/exceptions/Exception.c
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I./src -I./src -I./src/exceptions -I./src/lo -I./src/net -I./src/statistics -I./src/util -I./src
/thread -Wno-address -Wno-pointer-sign -g -O2 -D_REENTRANT -Wall -Wunused -Wno-unused-label -fsigned-char -D_GNU_SOURCE -std=c99 -c src/e
xceptions/Exception.c -fPIC -DPIC -o src/exceptions/.libs/Exception.o
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I./src -I./src -I./src/exceptions -I./src/lo -I./src/net -I./src/statistics -I./src/util -I./src
/thread -Wno-address -Wno-pointer-sign -g -O2 -D_REENTRANT -Wall -Wunused -Wno-unused-label -fsigned-char -D_GNU_SOURCE -std=c99 -c src/e
xceptions/Exception.c -o src/exceptions/Exception.o >/dev/null 2>&1
/bin/sh ./libtool --tagCC --mode=compile gcc -DHAVE_CONFIG_H -I. -I./src -I./src -I./src/exceptions -I./src/lo -I./src/net -I./src/st
statistics -I./src/util -I./src/thread -Wno-address -Wno-pointer-sign -g -O2 -D_REENTRANT -Wall -Wunused -Wno-unused-label -fsigned-char
-D_GNU_SOURCE -std=c99 -c -o src/lo/Dir.lo src/lo/Dir.c
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I./src -I./src -I./src/exceptions -I./src/lo -I./src/net -I./src/statistics -I./src/util -I./src
/thread -Wno-address -Wno-pointer-sign -g -O2 -D_REENTRANT -Wall -Wunused -Wno-unused-label -fsigned-char -D_GNU_SOURCE -std=c99 -c src/l
o/Dir.c -fPIC -DPIC -o src/lo/.libs/Dir.o
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I./src -I./src -I./src/exceptions -I./src/lo -I./src/net -I./src/statistics -I./src/util -I./src
/thread -Wno-address -Wno-pointer-sign -g -O2 -D_REENTRANT -Wall -Wunused -Wno-unused-label -fsigned-char -D_GNU_SOURCE -std=c99 -c src/l
o/Dir.c -o src/lo/Dir.o >/dev/null 2>&1
/bin/sh ./libtool --tagCC --mode=compile gcc -DHAVE_CONFIG_H -I. -I./src -I./src -I./src/exceptions -I./src/lo -I./src/net -I./src/st
statistics -I./src/util -I./src/thread -Wno-address -Wno-pointer-sign -g -O2 -D_REENTRANT -Wall -Wunused -Wno-unused-label -fsigned-char
```

```
CentOS 7 64-bit - VMware Workstation
File Edit View VM Tabs Help
CentOS 7 64-bit
Applications Places Terminal
trinpc192008@localhost:~/monit-5.25.2

File Edit View Search Terminal Help
[root@localhost monit-5.25.2]# sudo make install
make install-recursive
make[1]: Entering directory '/root/monit-5.25.2'
Making install in libmonit
make[2]: Entering directory '/root/monit-5.25.2/libmonit'
Making install in .
make[3]: Entering directory '/root/monit-5.25.2/libmonit'
make[4]: Entering directory '/root/monit-5.25.2/libmonit'
make[4]: Nothing to be done for 'install-exec-am'.
make[4]: Nothing to be done for 'install-data-am'.
make[4]: Leaving directory '/root/monit-5.25.2/libmonit'
make[3]: Leaving directory '/root/monit-5.25.2/libmonit'
Making install in test
make[3]: Entering directory '/root/monit-5.25.2/libmonit/test'
make[4]: Entering directory '/root/monit-5.25.2/libmonit/test'
make[4]: Nothing to be done for 'install-exec-am'.
make[4]: Nothing to be done for 'install-data-am'.
make[4]: Leaving directory '/root/monit-5.25.2/libmonit/test'
make[3]: Leaving directory '/root/monit-5.25.2/libmonit/test'
make[2]: Leaving directory '/root/monit-5.25.2/libmonit'
make[2]: Entering directory '/root/monit-5.25.2'
make[3]: Entering directory '/root/monit-5.25.2'
./usr/bin/mkdir -p '/usr/local/bin'
libtool: install: ./usr/bin/install -c monit '/usr/local/bin'
./usr/bin/mkdir -p '/usr/local/share/man/man1'
./usr/bin/install -c -m 644 monit.1 '/usr/local/share/man/man1'
make[3]: Leaving directory '/root/monit-5.25.2'
make[2]: Leaving directory '/root/monit-5.25.2'
make[1]: Leaving directory '/root/monit-5.25.2'
[root@localhost monit-5.25.2]# which monit
/usr/local/bin/monit
[root@localhost monit-5.25.2]# monit -V
This is Monit version 5.25.2
Built with ssl, with ipv6, with compression, with pam and with large files
Copyright (C) 2001-2018 Tildeslash Ltd. All Rights Reserved.
[root@localhost monit-5.25.2]#
```

***Note:** Install some libraries and tools before running the command **./configure**

sudo yum install gcc make openssl-devel -y

Install compiler and SSL development libraries

sudo yum install autoconf automake libtool -y

Install tools required for building from source

If error: libz not found

Fix: **sudo yum install zlib-devel -y** *# Install zlib compression library*

If error: PAM enabled but headers or library not found

Fix: **sudo yum install pam-devel -y** *# Install PAM authentication library*

If error: Couldn't find your SSL header files

Fix: **sudo yum install openssl-devel -y** *# Reinstall OpenSSL development headers*

2. Basic Configuration

a. Email Alert Configuration

Step 1:

sudo nano /etc/monitrc

Open the Monit configuration file for editing

Sample Configuration (including email alert). Find and edit the following lines

set daemon 30 *# Monit checks all services every 30 seconds*

Comment set log syslog or delete it and add:

set logfile /var/log/monit.log *# Log Monit activity to this file*

```
##
#####
## Global section
#####
##
## Start Monit in the background (run as a daemon):
#
set daemon 30          # check services at 30 seconds intervals
#   with start delay 240  # optional: delay the first check by 4-minutes (by
#                         # default Monit check immediately after Monit start)
#
#
## Set syslog logging. If you want to log to a standalone log file instead,
## specify the full path to the log file
#
#set log syslog
set logfile /var/log/monit.log
#
#
## Set the location of the Monit lock file which stores the process id of the

^G Get Help      ^O WriteOut      ^R Read File     ^Y Prev Page     ^K Cut Text       ^C Cur Pos
^X Exit          ^J Justify       ^W Where Is      ^V Next Page     ^U UnCut Text     ^T To Spell

trinpc192008@localhost:~ Problem loading page - Mozilla ... 1 / 4
```

Configure Gmail SMTP server for sending email alerts, add the following lines:

set mailserver smtp.gmail.com port 587

username "<your-email>@gmail.com" password "<your-app-password>"

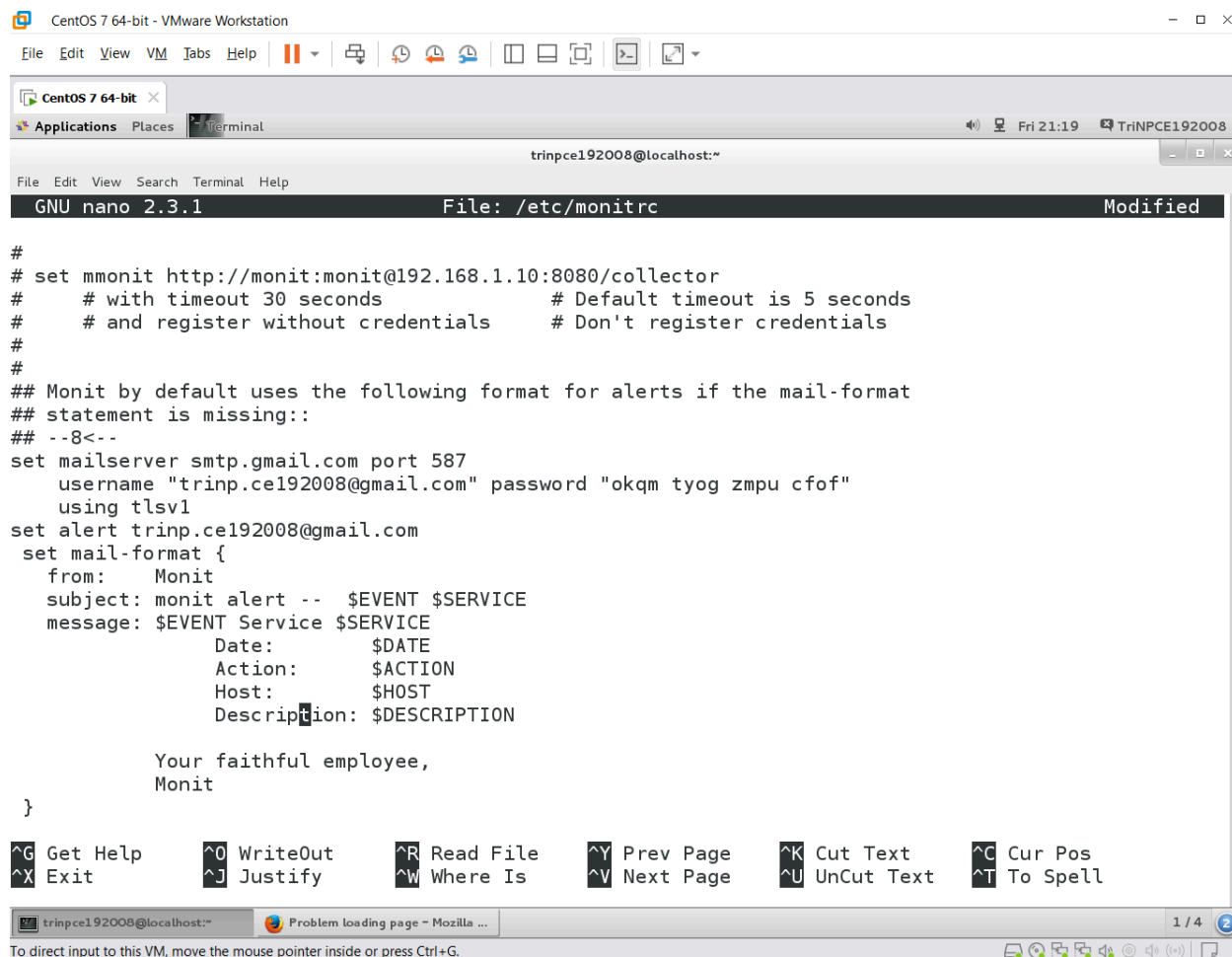
using tlsv1

set alert <your-email>@gmail.com *# Email address to receive alerts*

Customize the format of alert emails (subject and message body), find and delete comments (#) on lines:

set mail-format

```
{
  from: Monit
  subject: Monit Alert -- $EVENT $SERVICE
  message: Monit $ACTION $SERVICE at $DATE on $HOST: $DESCRIPTION.
}
```



```
#
# set mmonit http://monit:monit@192.168.1.10:8080/collector
#   # with timeout 30 seconds           # Default timeout is 5 seconds
#   # and register without credentials   # Don't register credentials
#
## Monit by default uses the following format for alerts if the mail-format
## statement is missing::
## --8<--
set mailserver smtp.gmail.com port 587
    username "trinp.ce192008@gmail.com" password "okqm tyog zmpu cfof"
    using tlsv1
set alert trinp.ce192008@gmail.com
set mail-format {
    from:      Monit
    subject:   monit alert -- $EVENT $SERVICE
    message:   $EVENT Service $SERVICE
               Date:      $DATE
               Action:     $ACTION
               Host:       $HOST
               Description: $DESCRIPTION

    Your faithful employee,
    Monit
}

^G Get Help      ^O WriteOut      ^R Read File     ^Y Prev Page     ^K Cut Text      ^C Cur Pos
^X Exit          ^J Justify       ^W Where Is      ^V Next Page     ^U UnCut Text    ^T To Spell
```

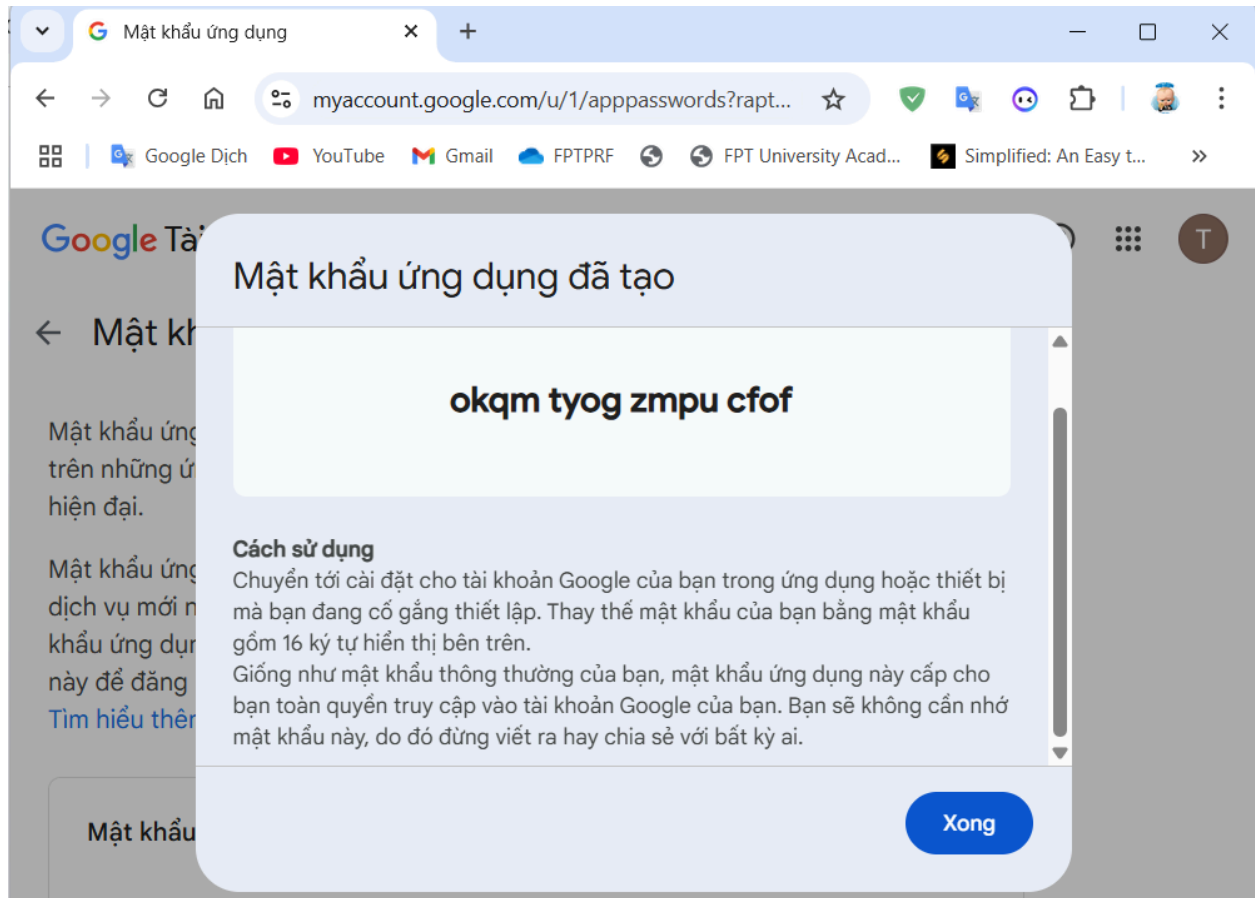
*** Note:** Replace placeholders

<your-email>: your Gmail address

<your-app-password>: generated via <https://myaccount.google.com/apppasswords>.

Enter the application name as Monit, then select Create Gmail will give you a code which is <your-app-password>.

Ensure that your Gmail account has 2-step verification enabled and that you are using a valid App Password generated from your Google account settings.



Step 2: Secure the `monitrc` File

```
sudo chmod 600 /etc/monitrc
```

Step 3: Run Monit with Custom Config

```
sudo systemctl start monit
```

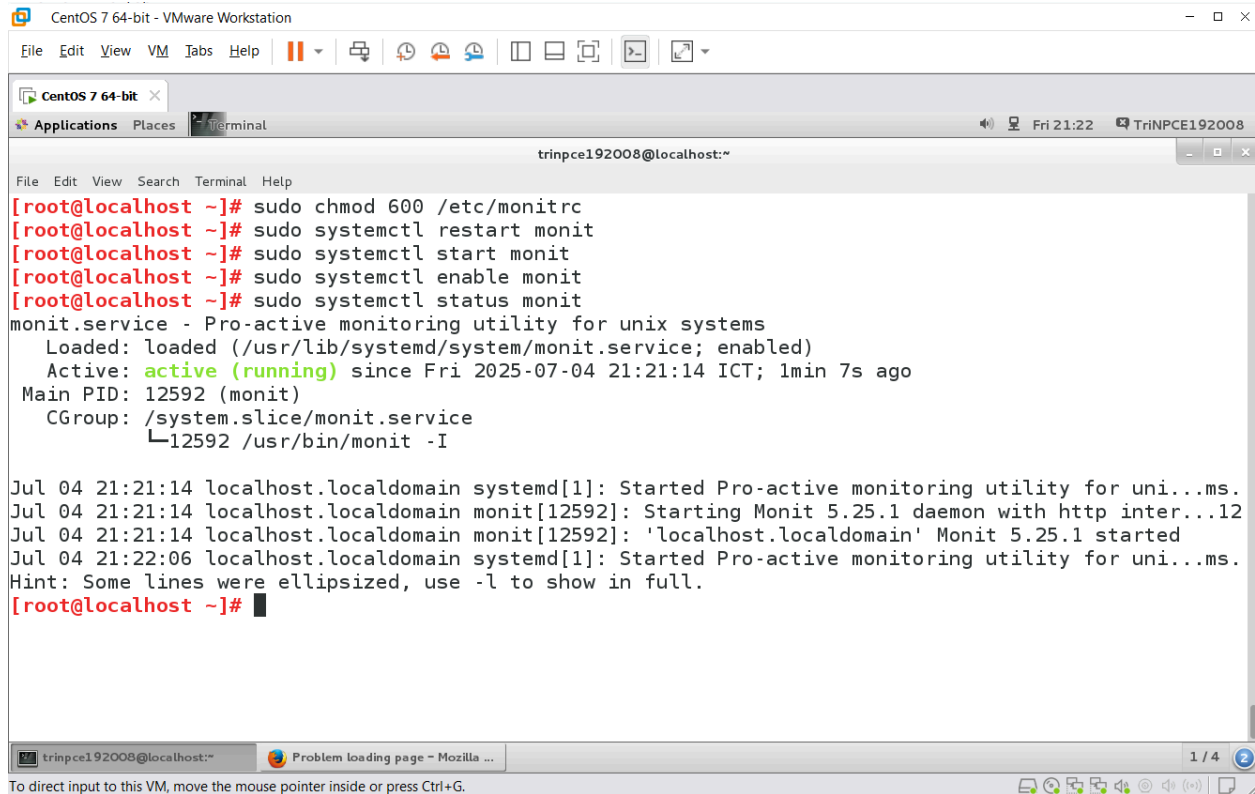
#Start Monit

```
sudo systemctl enable monit
```

#Enable Monit to start on boot

```
sudo systemctl status monit
```

#check Monit status make sure it is running



```
CentOS 7 64-bit - VMware Workstation
File Edit View VM Tabs Help
CentOS 7 64-bit x
Applications Places Terminal
trinpc192008@localhost:~
File Edit View Search Terminal Help
[root@localhost ~]# sudo chmod 600 /etc/monitrc
[root@localhost ~]# sudo systemctl restart monit
[root@localhost ~]# sudo systemctl start monit
[root@localhost ~]# sudo systemctl enable monit
[root@localhost ~]# sudo systemctl status monit
monit.service - Pro-active monitoring utility for unix systems
Loaded: loaded (/usr/lib/systemd/system/monit.service; enabled)
Active: active (running) since Fri 2025-07-04 21:21:14 ICT; 1min 7s ago
Main PID: 12592 (monit)
CGroup: /system.slice/monit.service
└─12592 /usr/bin/monit -I

Jul 04 21:21:14 localhost.localdomain systemd[1]: Started Pro-active monitoring utility for uni...ms.
Jul 04 21:21:14 localhost.localdomain monit[12592]: Starting Monit 5.25.1 daemon with http inter...12
Jul 04 21:21:14 localhost.localdomain monit[12592]: 'localhost.localdomain' Monit 5.25.1 started
Jul 04 21:22:06 localhost.localdomain systemd[1]: Started Pro-active monitoring utility for uni...ms.
Hint: Some lines were ellipsized, use -l to show in full.
[root@localhost ~]#
```

Step 4: Check the email to view the alert sent by Monit.

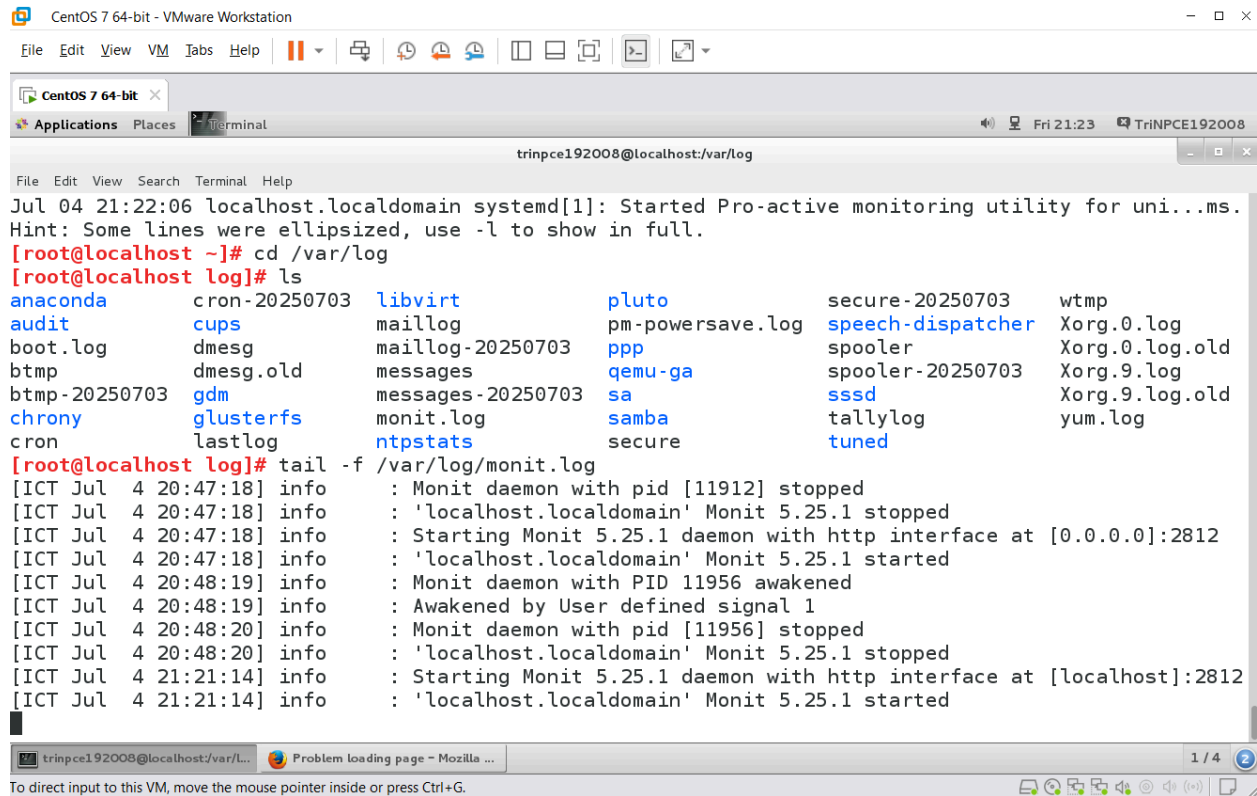
***Note:** Avoid using personal email addresses, as Monit logs detailed system information that may increase the risk of spam or data exposure.

b. Log and Web Interface Configuration

i. View Logs in file

tail -f /var/log/monit.log

- The log file contains all actions Monit performs: restart, alert, error...
- **tail -f**: realtime monitoring when Monit writes new logs



```
CentOS 7 64-bit - VMware Workstation
File Edit View VM Tabs Help
CentOS 7 64-bit x
Applications Places Terminal
trinpc192008@localhost:/var/log
File Edit View Search Terminal Help
Jul 04 21:22:06 localhost.localdomain systemd[1]: Started Pro-active monitoring utility for uni...ms.
Hint: Some lines were ellipsized, use -l to show in full.
[root@localhost ~]# cd /var/log
[root@localhost log]# ls
anaconda      cron-20250703  libvirt        pluto          secure-20250703  wtmp
audit         cups          maillog        pm-powersave.log speech-dispatcher Xorg.0.log
boot.log      dmesg         maillog-20250703  ppp            spooler          Xorg.0.log.old
btmp         dmesg.old     messages       qemu-ga        spooler-20250703 Xorg.9.log
btmp-20250703 gdm           messages-20250703 sa             sssd            Xorg.9.log.old
chrony        glusterfs     monit.log       samba          tallylog         yum.log
cron          lastlog       ntpstats       secure         tuned
[root@localhost log]# tail -f /var/log/monit.log
[ICT Jul  4 20:47:18] info      : Monit daemon with pid [11912] stopped
[ICT Jul  4 20:47:18] info      : 'localhost.localdomain' Monit 5.25.1 stopped
[ICT Jul  4 20:47:18] info      : Starting Monit 5.25.1 daemon with http interface at [0.0.0.0]:2812
[ICT Jul  4 20:47:18] info      : 'localhost.localdomain' Monit 5.25.1 started
[ICT Jul  4 20:48:19] info      : Monit daemon with PID 11956 awakened
[ICT Jul  4 20:48:19] info      : Awakened by User defined signal 1
[ICT Jul  4 20:48:20] info      : Monit daemon with pid [11956] stopped
[ICT Jul  4 20:48:20] info      : 'localhost.localdomain' Monit 5.25.1 stopped
[ICT Jul  4 21:21:14] info      : Starting Monit 5.25.1 daemon with http interface at [localhost]:2812
[ICT Jul  4 21:21:14] info      : 'localhost.localdomain' Monit 5.25.1 started
```

ii. Web interface management

Step 1: Open Monit configuration file

sudo nano /etc/monitrc

Step 2:

To facilitate remote monitoring and administration of the Monit service, the configuration should be modified to allow external connections, find, delete comments (#) and edit lines:

set httpd port 2812 and

use address 0.0.0.0

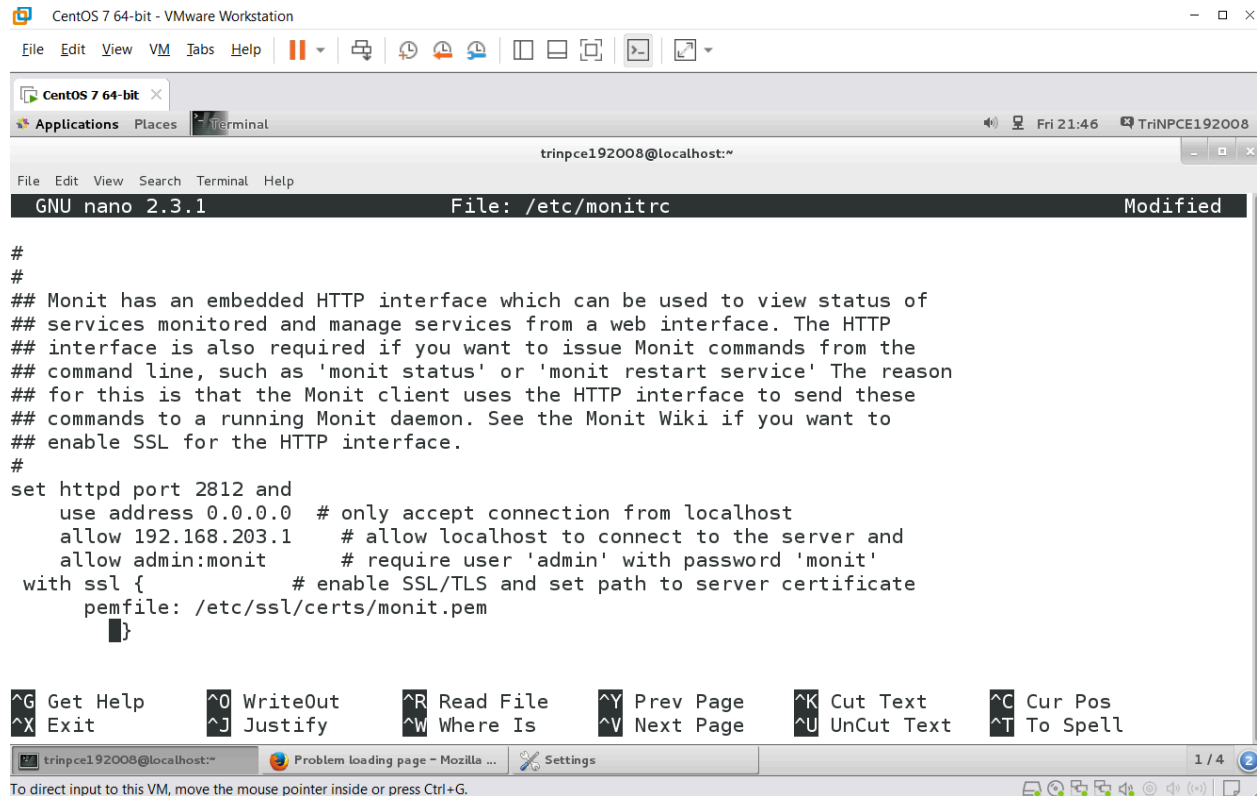
Accept connections from any IP address

```
allow <your-ip-address> # Replace with the IP address of the remote machine
allow admin:monit       # Set access credentials (username, password to enter Monit)
```

enable SSL/TLS and set path to server certificate

with ssl

```
{pemfile: /etc/ssl/certs/monit.pem }
```



The screenshot shows a terminal window titled 'CentOS 7 64-bit - VMware Workstation'. The terminal is running the nano text editor, editing the file '/etc/monitrc'. The content of the file is as follows:

```
#
#
## Monit has an embedded HTTP interface which can be used to view status of
## services monitored and manage services from a web interface. The HTTP
## interface is also required if you want to issue Monit commands from the
## command line, such as 'monit status' or 'monit restart service' The reason
## for this is that the Monit client uses the HTTP interface to send these
## commands to a running Monit daemon. See the Monit Wiki if you want to
## enable SSL for the HTTP interface.
#
set httpd port 2812 and
  use address 0.0.0.0 # only accept connection from localhost
  allow 192.168.203.1 # allow localhost to connect to the server and
  allow admin:monit  # require user 'admin' with password 'monit'
with ssl {           # enable SSL/TLS and set path to server certificate
  pemfile: /etc/ssl/certs/monit.pem
}
```

The terminal window also shows the nano editor's status bar at the bottom with various shortcuts like ^G Get Help, ^X Exit, ^O WriteOut, etc.

***Note:**

*To permit access from multiple sources, administrators should append multiple **allow** directives. Example:*

```
allow 192.168.11.78    # Allow a specific IP address
allow 192.168.1.20     # Allow another specific IP address
allow 192.168.1.0/24   # Allow all devices within a subnet
```

To restrict access to Monit's web interface to the local server only to enhance security by preventing remote connections, configure as follows:

set httpd port 2812 and

use address localhost

allow localhost

allow admin:monit

```
set httpd port 2812 and
  use address localhost # only accept connection from localhost
  allow localhost      # allow localhost to connect to the server and
  allow admin:monit    # require user 'admin' with password 'monit'
```

Step 3:

To allow remote access to Monit's web interface, configure the firewall to permit inbound traffic on port 2812 by adding a rule to **iptables**:

sudo iptables -I INPUT -p tcp --dport 2812 -j ACCEPT

Allow incoming HTTP connections to Monit's web interface (port 2812)

sudo iptables-save > /etc/sysconfig/iptables

Save the current firewall rules so they persist after reboot

sudo systemctl status iptables *# Check if the iptables service is active and running*

sudo systemctl enable iptables *#Enable Iptable if it is not running*

sudo systemctl start iptables *#Enable Iptable to start with system*

*#If using **firewalld** (CentOS 7 and newer):*

sudo firewall-cmd --permanent --add-port=2812/tcp

sudo firewall-cmd --reload

sudo systemctl restart monit *#Restart Monit to update configuration*

```
CentOS 7 64-bit - VMware Workstation
File Edit View VM Tabs Help
CentOS 7 64-bit x
Applications Places Terminal
trinpc192008@localhost:~
File Edit View Search Terminal Help
[tr@localhost ~]$ sudo iptables -I INPUT -p tcp --dport 2812 -j ACCEPT
[tr@localhost ~]$ sudo iptables-save > /etc/sysconfig/iptables
[tr@localhost ~]$ sudo systemctl enable iptables
ln -s '/usr/lib/systemd/system/iptables.service' '/etc/systemd/system/basic.target.wants/iptables.service'
[tr@localhost ~]$ sudo systemctl start iptables
[tr@localhost ~]$ sudo systemctl status iptables
iptables.service - IPv4 firewall with iptables
  Loaded: loaded (/usr/lib/systemd/system/iptables.service; enabled)
  Active: active (exited) since Fri 2025-07-04 20:02:50 ICT; 1h 29min ago
  Main PID: 10430 (code=exited, status=0/SUCCESS)
  CGroup: /system.slice/iptables.service

Jul 04 20:02:50 localhost.localdomain systemd[1]: Starting IPv4 firewall with iptables...
Jul 04 20:02:50 localhost.localdomain iptables.init[10430]: iptables: Applying firewall rules: [...]
Jul 04 20:02:50 localhost.localdomain systemd[1]: Started IPv4 firewall with iptables.
Jul 04 20:11:53 localhost.localdomain systemd[1]: Started IPv4 firewall with iptables.
Jul 04 21:32:01 localhost.localdomain systemd[1]: Started IPv4 firewall with iptables.
Hint: Some lines were ellipsized, use -l to show in full.
[tr@localhost ~]$ sudo systemctl restart monit
[tr@localhost ~]$
```

Step 4:

On local server, open <http://localhost:2812/>, input username *admin* password *monit*.

From another computer, access <http://<IP-Server>:2812>

Example: <http://192.168.203.131:2812>

Monit: localhost.localdomain

Không bảo mật 192.168.203.131:2812

Google Dịch YouTube Gmail FPTPRF FPT University Acad... Simplified: An Easy t... Microsoft Designer...

Home > Use M/Monit to manage all your Monit instances Monit 5.25.1

Monit Service Manager

Monit is running on localhost.localdomain and monitoring:

System	Status	Load	CPU	Memory	Swap
localhost.localdomain	OK	[0.05] [0.10] [0.13]	2.0%us, 0.3%sy, 0.0%wa	70.9% [695.1 MB]	14.5% [297.8 MB]

Step 5: To enable Monit to manage a service, a configuration file must be placed in **/etc/monit.d/**

Additionally, the main configuration file **/etc/monitrc** must include the following directive to ensure that Monit loads all service definitions from that directory:

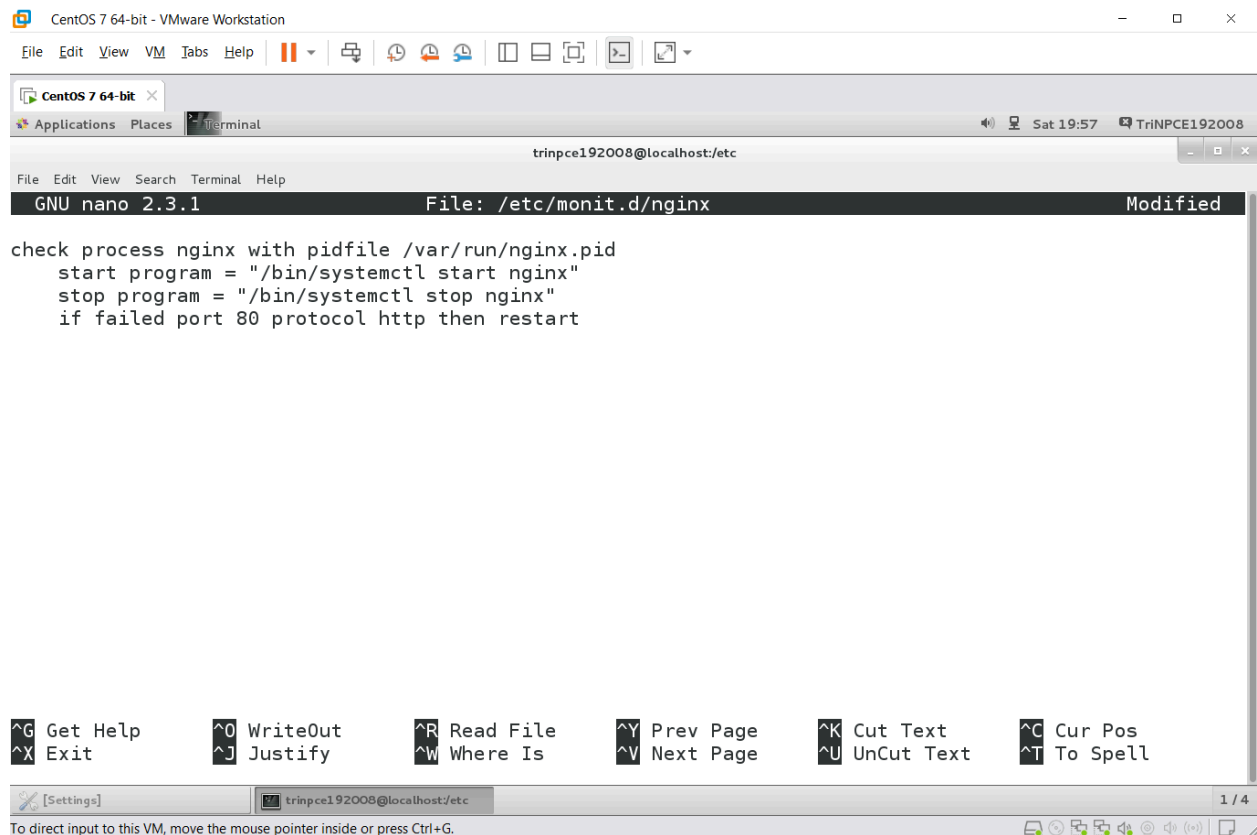
include /etc/monit.d/*

This directive instructs Monit to read and apply all individual configuration files stored in **/etc/monit.d/**, allowing modular service monitoring.

Example: set up monitoring for the Nginx service

#Create file /etc/monit.d/nginx

sudo nano /etc/monit.d/nginx



```
CentOS 7 64-bit - VMware Workstation
File Edit View VM Tabs Help
CentOS 7 64-bit x
Applications Places Terminal
trinpc192008@localhost:/etc
File Edit View Search Terminal Help
GNU nano 2.3.1 File: /etc/monit.d/nginx Modified
check process nginx with pidfile /var/run/nginx.pid
start program = "/bin/systemctl start nginx"
stop program = "/bin/systemctl stop nginx"
if failed port 80 protocol http then restart

^G Get Help      ^O WriteOut      ^R Read File     ^Y Prev Page     ^K Cut Text      ^C Cur Pos
^X Exit          ^J Justify       ^W Where Is     ^V Next Page     ^U UnCut Text    ^T To Spell

[Settings] trinpc192008@localhost/etc 1 / 4
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.
```

```
CentOS 7 64-bit - VMware Workstation
File Edit View VM Tabs Help
CentOS 7 64-bit
Applications Places Terminal
trinpce192008@localhost/etc
File Edit View Search Terminal Help
[root@localhost etc]# nano /etc/monit.d/nginx
[root@localhost etc]# sudo systemctl restart monit
[root@localhost etc]# sudo systemctl status monit
monit.service - Pro-active monitoring utility for unix systems
  Loaded: loaded (/usr/lib/systemd/system/monit.service; enabled)
  Active: active (running) since Sat 2025-07-05 19:57:48 ICT; 20s ago
  Process: 9124 ExecStop=/usr/bin/monit quit (code=exited, status=0/SUCCESS)
  Main PID: 9126 (monit)
  CGroup: /system.slice/monit.service
          └─9126 /usr/bin/monit -I

Jul 05 19:57:48 localhost.localdomain systemd[1]: Starting Pro-active monitoring utility for unix .....
Jul 05 19:57:48 localhost.localdomain systemd[1]: Started Pro-active monitoring utility for unix s...ms.
Jul 05 19:57:48 localhost.localdomain monit[9126]: Starting Monit 5.25.1 daemon with http interfac...812
Jul 05 19:57:48 localhost.localdomain monit[9126]: 'localhost.localdomain' Monit 5.25.1 started
Hint: Some lines were ellipsized, use -l to show in full.
```

After reloading the Monit web interface, detailed information about the Nginx service will be displayed. By clicking on the "nginx" entry, users can access available control actions such as **start**, **stop**, or **restart** the service:

← → ↺ 🏠 ⚠️ Không bảo mật 192.168.203.135:2812 🔍 🌟 🛡️ 🗨️ 🔄 📄 🧑🏻

📦 | 🌐 Google Dịch 📺 YouTube 📧 Gmail ☁️ FPTPRF 🌐 🌐 FPT University Acad... 📱 Simplified: An Easy t... 🎨 Microsoft Designer... 🗨️ Writesonic - AI Cont... >>

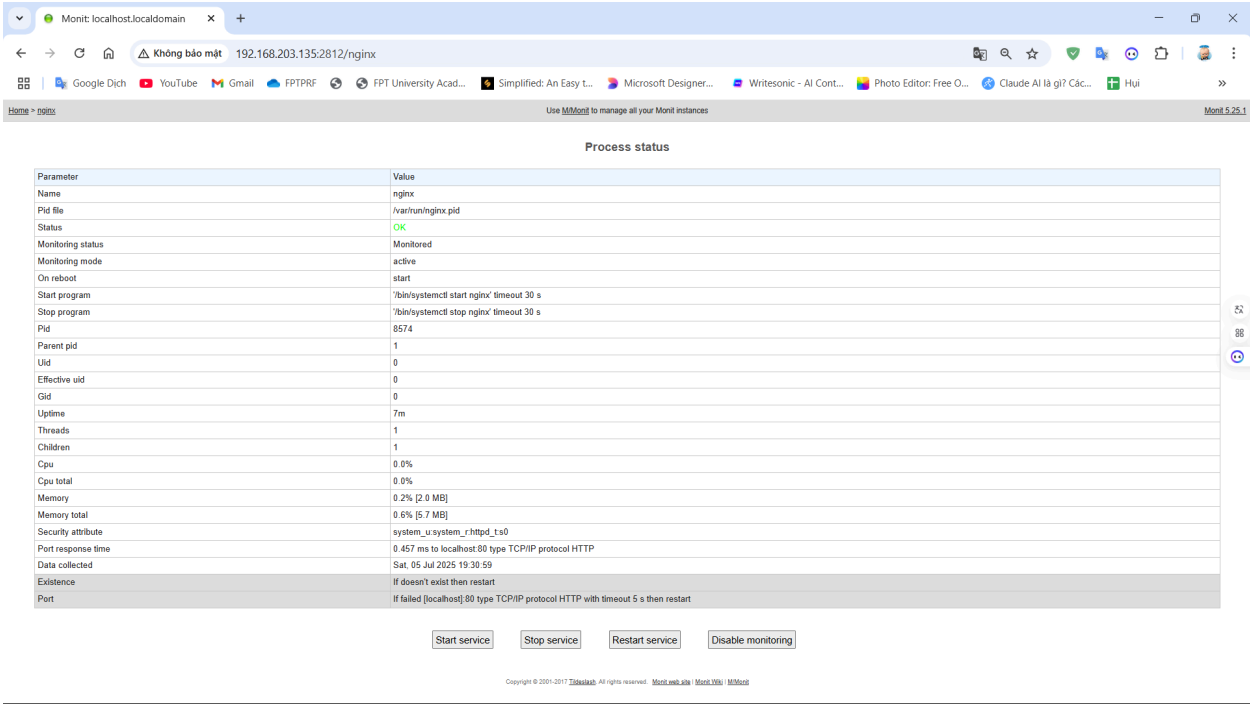
Home > Use **MMonit** to manage all your Monit instances Monit 5.25

Monit Service Manager

Monit is **running** on localhost.localdomain and monitoring:

System	Status	Load	CPU	Memory	Swap
localhost.localdomain	OK	[0.01] [0.08] [0.09]	0.4%us, 0.2%sy, 0.0%wa	71.4% [699.8 MB]	13.2% [269.9 MB]

Process	Status	Uptime	CPU Total	Memory Total	Read	Write
nginx	OK	2m	0.0%	0.6% [5.7 MB]	-	-



III. Advanced Monitoring Configuration

1. Process Monitoring

a. Syntax Specification

CHECK PROCESS *<unique name>* **PIDFILE** *<path>* | **MATCHING** *<regex>*

Description of Parameters:

CHECK PROCESS: Keyword used to define process monitoring in Monit.

<unique name>: A unique identifier for the process being monitored.

PIDFILE *<path>*: Specifies the path to the PID (Process ID) file of the process. This allows Monit to track the exact running instance.

MATCHING *<regex>*: Uses a regular expression to match the process name or attributes from the process list.

|: Indicates that you must choose either **PIDFILE** or **MATCHING**, but not both at the same time.

b. Example Configuration

i. Sample Configuration – Monitoring with PIDFILE

Step 1: Define a new service for Monit to monitor

```
yum install nginx           #install nginx process.
```

```
sudo systemctl start nginx #to start nginx.
```

#Next, configure Monit to supervise the Nginx service by creating a configuration file.

```
sudo vim /etc/monit.d/nginx-monitor
```

#Insert the following content

```
check process nginx with pidfile /run/nginx.pid
```

```
start program "/usr/bin/systemctl start nginx.service"
```

```
stop program "/usr/bin/systemctl stop nginx.service"
```

```
if failed port 80 protocol http then restart
```

```
alert quocbaoduong231@gmail.com
```

After saving the configuration file, exit the editor.

```
File Edit View Search Terminal Help
[root@localhost log]# yum install nginx
Loaded plugins: fastestmirror, langpacks
Loading mirror speeds from cached hostfile
* epel: d2lzk17pfhq30w.cloudfront.net
Package 1:nginx-1.20.1-10.el7.x86_64 already installed and latest version
Nothing to do
[root@localhost log]# sudo systemctl start nginx
[root@localhost log]# █
```

```
Home X CentOS 7 64-bit (3) X
Applications Places Terminal
File Edit View Search Terminal Help
check process nginx with pidfile /run/nginx.pid
start program "/usr/bin/systemctl start nginx.service"
stop program "/usr/bin/systemctl stop nginx.service"
if failed port 80 protocol http then restart
alert quocbaoduong231@gmail.com
~
```

***Note:**

Ensure that the line **set alert** <your-email>**@gmail.com** is included in the Monit configuration file. Without this line, Monit will not be able to send alert notifications via email.

```
sudo monit reload    # Reloads Monit with updated configuration.
sudo monit status   # Displays the current status of all monitored services.
```

Step 3: Stop the Nginx service to trigger a Monit alert

```
sudo systemctl stop nginx    # Stops the Nginx service.
sudo systemctl status nginx  # Verifies that Nginx is inactive.
```

```
[root@localhost log]# systemctl stop nginx
[root@localhost log]# systemctl status nginx
● nginx.service - The nginx HTTP and reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; vendor preset: disabled)
   Active: inactive (dead) since Sun 2025-07-06 08:33:07 PDT; 10s ago
     Process: 18433 ExecStart=/usr/sbin/nginx (code=exited, status=0/SUCCESS)
     Process: 18430 ExecStartPre=/usr/sbin/nginx -t (code=exited, status=0/SUCCESS)
     Process: 18427 ExecStartPre=/usr/bin/rm -f /run/nginx.pid (code=exited, status=0/SUCCESS)
    Main PID: 18436 (code=exited, status=0/SUCCESS)

Jul 06 06:53:33 localhost.localdomain systemd[1]: Starting The nginx HTTP and reverse proxy server...
Jul 06 06:53:33 localhost.localdomain nginx[18430]: nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
Jul 06 06:53:33 localhost.localdomain nginx[18430]: nginx: configuration file /etc/nginx/nginx.conf test is successful
Jul 06 06:53:33 localhost.localdomain systemd[1]: Started The nginx HTTP and reverse proxy server.
Jul 06 08:33:07 localhost.localdomain systemd[1]: Stopping The nginx HTTP and reverse proxy server...
Jul 06 08:33:07 localhost.localdomain systemd[1]: Stopped The nginx HTTP and reverse proxy server.
[root@localhost log]# █
```

Step 4: Monitor Monit's log file in real-time to check if an alert is triggered

sudo tail -f /var/log/monit.log

```
[root@localhost log]# sudo tail -f /var/log/monit.log
[2025-07-06T06:54:04-0700] error : Mail: Authentication failed -- no supported authentication methods found
[2025-07-06T06:54:04-0700] error : Aborting event
[2025-07-06T08:33:16-0700] error : 'nginx' process is not running
[2025-07-06T08:33:17-0700] error : Mail: Authentication failed -- no supported authentication methods found
[2025-07-06T08:33:17-0700] error : Aborting event
[2025-07-06T08:33:17-0700] info : 'nginx' trying to restart
[2025-07-06T08:33:17-0700] info : 'nginx' start: '/usr/bin/systemctl start nginx.service'
[2025-07-06T08:33:47-0700] info : 'nginx' process is running with pid 20514
[2025-07-06T08:33:48-0700] error : Mail: Authentication failed -- no supported authentication methods found
[2025-07-06T08:33:49-0700] error : Aborting event
█
```

***Note:**

If you encounter the error message:

"Mail: Authentication failed -- no supported authentication methods found"

This issue is related to Google App Password misconfiguration, not Monit itself. Ensure that your Gmail account has 2-step verification enabled and that you are using a valid App Password generated from your Google account settings.

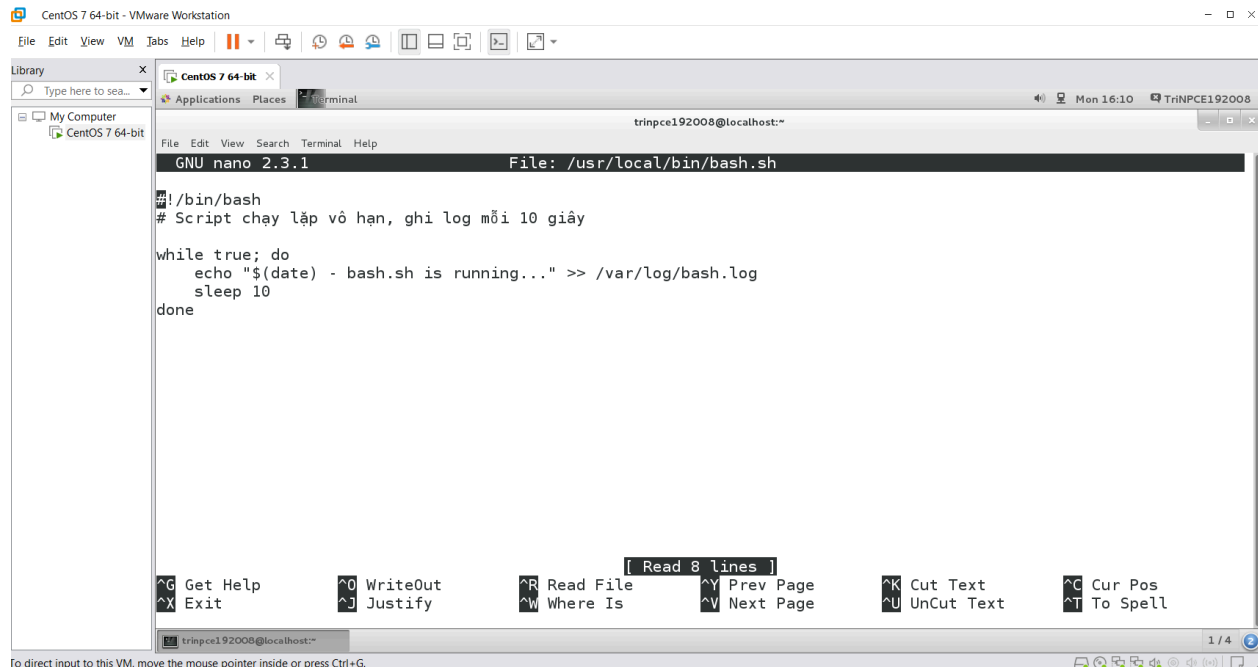
ii. Sample Configuration – Monitoring with MATCHING

*In some cases, the application does not create a PID file. In that case, Monit can use the MATCHING method to find the process based on the name displayed in the **ps** command.*

*Example with **bash.sh**.*

Step 1: Create a simple script, for example a logging loop:

sudo nano /usr/local/bin/bash.sh



```
#!/bin/bash
# Script chạy lặp vô hạn, ghi log mỗi 10 giây

while true; do
    echo "$(date) - bash.sh is running..." >> /var/log/bash.log
    sleep 10
done
```

#Decentralization of execution: **sudo chmod +x /usr/local/bin/bash.sh**

Step 2: Create configuration file

sudo nano /etc/monit.d/bash_script

#Insert the following content

*#Instructs Monit to find a running process that matches the string "bash.sh" from the output of **ps aux**.*

CHECK PROCESS bash_script MATCHING "bash.sh"

START PROGRAM = "/bin/bash /usr/local/bin/bash.sh &"

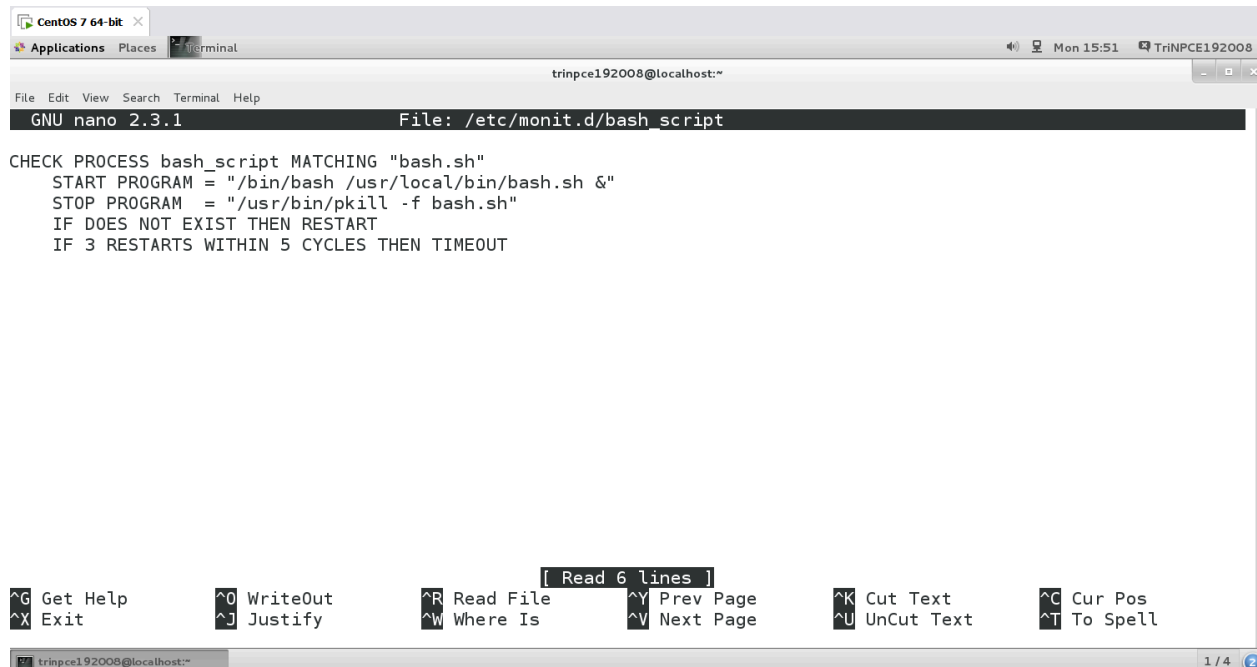
STOP PROGRAM = "/usr/bin/pkill -f bash.sh"

#Monit will automatically restart the process if it is not found.

IF DOES NOT EXIST THEN RESTART

#If Monit tries to restart the process 3 times within 5 cycles (check intervals), it will stop trying and consider it failed.

IF 3 RESTARTS WITHIN 5 CYCLES THEN TIMEOUT



```
CentOS 7 64-bit x
Applications Places terminal
trinpc192008@localhost:~
File Edit View Search Terminal Help
GNU nano 2.3.1 File: /etc/monit.d/bash_script

CHECK PROCESS bash_script MATCHING "bash.sh"
START PROGRAM = "/bin/bash /usr/local/bin/bash.sh &"
STOP PROGRAM = "/usr/bin/pkill -f bash.sh"
IF DOES NOT EXIST THEN RESTART
IF 3 RESTARTS WITHIN 5 CYCLES THEN TIMEOUT

[ Read 6 lines ]
^G Get Help      ^O WriteOut      ^R Read File     ^Y Prev Page     ^K Cut Text      ^C Cur Pos
^X Exit          ^J Justify       ^W Where Is      ^V Next Page     ^U UnCut Text    ^T To Spell
trinpc192008@localhost:~ 1 / 4
```

Step 3: Validate and Apply the Configuration

```
sudo monit -t      # Check for syntax errors.
sudo monit reload # Reload Monit configuration.
sudo monit status # Verify the process is being monitored.
```

***Note:** Run the following command to view the full process list and search for your target:

```
ps aux | grep bash.sh
```

```
trinpcel92008@localhost:~  
File Edit View Search Terminal Help  
[root@localhost ~]# sudo monit -t  
Skipping 'allow 127.0.0.1' -- host resolved to [::ffff:127.0.0.1] which is present in ACL already  
Control file syntax OK  
[root@localhost ~]# sudo monit reload  
Skipping 'allow 127.0.0.1' -- host resolved to [::ffff:127.0.0.1] which is present in ACL already  
Reinitializing monit daemon  
[root@localhost ~]# sudo monit status  
Skipping 'allow 127.0.0.1' -- host resolved to [::ffff:127.0.0.1] which is present in ACL already  
Monit 5.25.1 uptime: 7m  
  
Process 'nginx'  
status OK  
monitoring status Monitored  
monitoring mode active  
on reboot start  
pid 1617  
parent pid 1  
uid 0  
effective uid 0  
gid 0  
uptime 45m  
threads 1  
children 1  
cpu -  
cpu total -  
  
trinpcel92008@localhost:~ 1 / 4  
  
Applications Places Terminal  
trinpcel92008@localhost:~  
File Edit View Search Terminal Help  
Process 'bash_script'  
status OK  
monitoring status Monitored  
monitoring mode active  
on reboot start  
pid 4174  
parent pid 3681  
uid 0  
effective uid 0  
gid 0  
uptime 20m  
threads 1  
children 1  
cpu 0.0%  
cpu total 0.0%  
memory 0.1% [1.4 MB]  
memory total 0.2% [1.8 MB]  
security attribute unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023  
disk read 0 B/s [128 kB total]  
disk write 0 B/s [128 kB total]  
data collected Mon, 14 Jul 2025 15:42:25  
  
System 'localhost.localdomain'  
status OK  
  
trinpcel92008@localhost:~ 1 / 4  
ove the mouse pointer outside or press Ctrl+Alt.
```

2. File, Directory, and Filesystem Monitoring

a. Syntax Specification

CHECK FILE <unique name> **PATH** <path>

CHECK FILESYSTEM <unique name> **PATH** <string>

CHECK DIRECTORY <unique name> **PATH** <path>

Description of Parameters:

CHECK FILE: Monitor a specific file (existence, size, permissions...).

CHECK FILESYSTEM: Monitor a mounted filesystem (e.g., disk usage).

CHECK DIRECTORY: Monitor a folder for changes (permissions, ownership...).

<unique_name>: A unique name for this check.

PATH: Full path to the file, directory, or mount point (e.g., /, /home, /var).

b. Example Configuration

Multiple Monit configuration files can be created to monitor different files or mount points as needed.

Paths such as /etc/ssh/ssh_d_config and / are illustrative and should be replaced with those relevant to the system.

i. Sample Configuration – Monitoring Files

Step 1: Create a Configuration File

sudo nano /etc/monit.d/ssh_config_check

***Note:** Example of **ssh_d.sh** file content for monitoring:

sudo nano /usr/local/bin/check_ssh_d.sh #create file **ssh_d.sh**

#Add the following configuration

#!/bin/bash

FILE="/etc/ssh/ssh_d_config"

if [! -f "\$FILE"]; then

echo "ssh_d_config missing"

exit 1

fi

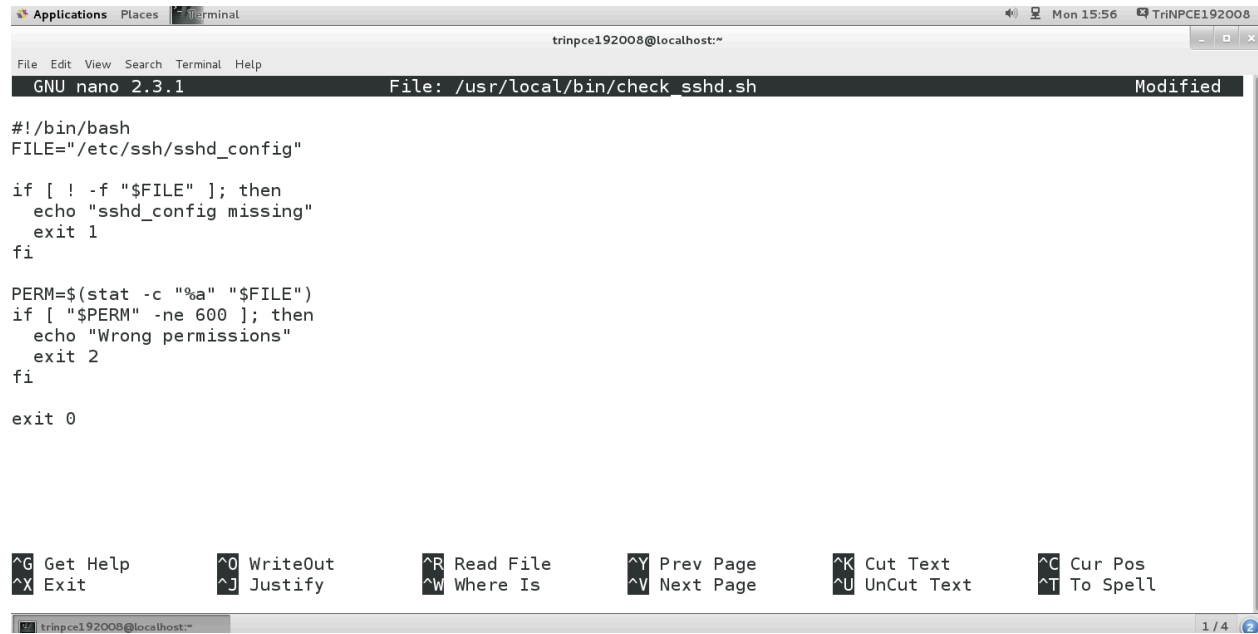
PERM=\$(stat -c "%a" "\$FILE")

if ["\$PERM" -ne 600]; then

echo "Wrong permissions"

exit 2
fi

exit 0



The screenshot shows a terminal window with the nano text editor open. The editor is editing a file named `/usr/local/bin/check_sshd.sh`. The script content is as follows:

```
#!/bin/bash
FILE="/etc/ssh/sshd_config"

if [ ! -f "$FILE" ]; then
    echo "sshd_config missing"
    exit 1
fi

PERM=$(stat -c "%a" "$FILE")
if [ "$PERM" -ne 600 ]; then
    echo "Wrong permissions"
    exit 2
fi

exit 0
```

The terminal window has a menu bar with options: File, Edit, View, Search, Terminal, Help. The status bar at the bottom shows the user `trinpc192008@localhost` and the page number `1 / 4`.

#Decentralization of execution: **sudo chmod +x /usr/local/bin/check_sshd.sh**

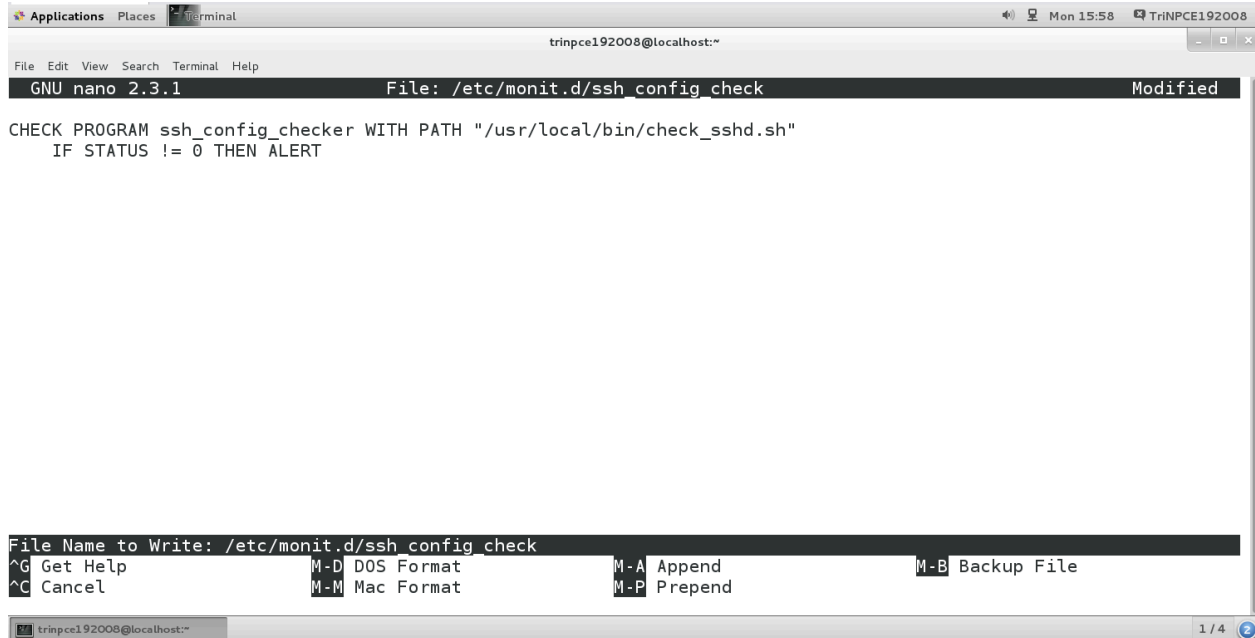
#Add the following configuration

#Monitors the SSH configuration file.

CHECK PROGRAM ssh_config_checker WITH PATH

"/usr/local/bin/check_sshd.sh"

IF STATUS != 0 THEN ALERT



```
Applications  Places  Terminal
trinpce192008@localhost:~
File Edit View Search Terminal Help
GNU nano 2.3.1 File: /etc/monit.d/ssh_config_check Modified

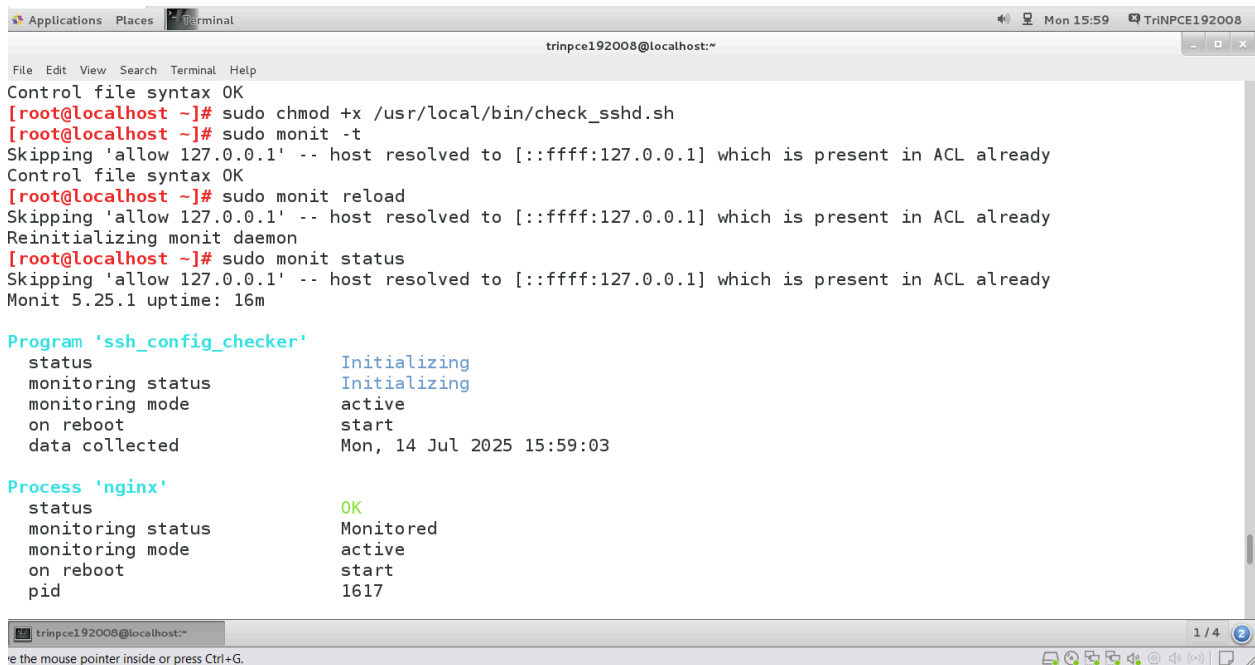
CHECK PROGRAM ssh_config_checker WITH PATH "/usr/local/bin/check_sshd.sh"
  IF STATUS != 0 THEN ALERT

File Name to Write: /etc/monit.d/ssh_config_check
^G Get Help      M-D DOS Format   M-A Append      M-B Backup File
^C Cancel        M-M Mac Format   M-P Prepend

trinpce192008@localhost:~ 1 / 4
```

Step 2: Reload Monit and Check Status

```
sudo monit -t
sudo monit reload
sudo monit status
```



```
Applications  Places  Terminal
trinpce192008@localhost:~
File Edit View Search Terminal Help
Control file syntax OK
[root@localhost ~]# sudo chmod +x /usr/local/bin/check_sshd.sh
[root@localhost ~]# sudo monit -t
Skipping 'allow 127.0.0.1' -- host resolved to [::ffff:127.0.0.1] which is present in ACL already
Control file syntax OK
[root@localhost ~]# sudo monit reload
Skipping 'allow 127.0.0.1' -- host resolved to [::ffff:127.0.0.1] which is present in ACL already
Reinitializing monit daemon
[root@localhost ~]# sudo monit status
Skipping 'allow 127.0.0.1' -- host resolved to [::ffff:127.0.0.1] which is present in ACL already
Monit 5.25.1 uptime: 16m

Program 'ssh_config_checker'
  status           Initializing
  monitoring status Initializing
  monitoring mode   active
  on reboot         start
  data collected    Mon, 14 Jul 2025 15:59:03

Process 'nginx'
  status           OK
  monitoring status Monitored
  monitoring mode   active
  on reboot         start
  pid              1617

trinpce192008@localhost:~ 1 / 4
e the mouse pointer inside or press Ctrl+G.
```

ii. Sample Configuration – Monitoring Filesystem

Step 1: Create a Configuration File

Example: monitor partition changes /

sudo nano /etc/monit.d/filesystem_root

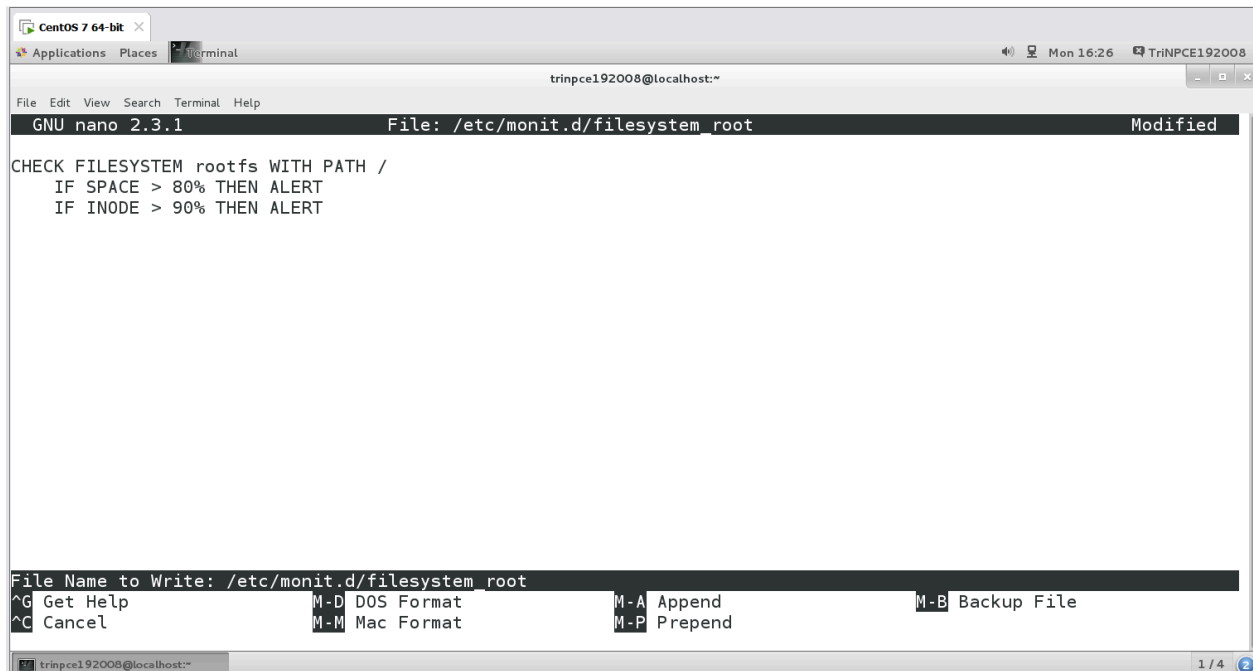
#Add the following configuration

CHECK FILESYSTEM rootfs WITH PATH / *#Monitors the root partition (/).*

#Sends alerts if disk space exceeds 80% or inode usage exceeds 90%.

IF SPACE > 80% THEN ALERT

IF INODE > 90% THEN ALERT



```
CentOS 7 64-bit
Applications Places Terminal Help
trinpcel92008@localhost:~
File Edit View Search Terminal Help
GNU nano 2.3.1 File: /etc/monit.d/filesystem_root Modified
CHECK FILESYSTEM rootfs WITH PATH /
IF SPACE > 80% THEN ALERT
IF INODE > 90% THEN ALERT
File Name to Write: /etc/monit.d/filesystem_root
^G Get Help      M-D DOS Format   M-A Append      M-B Backup File
^C Cancel        M-M Mac Format   M-P Prepend
trinpcel92008@localhost:~ 1 / 4
```

Step 2: Reload Monit and Check Status

sudo monit -t

sudo monit reload

sudo monit status

```
Applications Places Terminal
trinpc192008@localhost:~
File Edit View Search Terminal Help

Filesystem 'rootfs'
status OK
monitoring status Monitored
monitoring mode active
on reboot start
filesystem type xfs
filesystem flags rw,seclabel,relatime,attr2,inode64,noquota
permission 755
uid 0
gid 0
block size 4 kB
space total 10.8 GB (of which 0.0% is reserved for root user)
space free for non superuser 5.8 GB [53.9%]
space free total 5.8 GB [53.9%]
inodes total 11288576
inodes free 11173591 [99.0%]
read 0 B/s [2.0 GB total], 0.0 reads/s [81985 reads total]
write 0 B/s [75.0 MB total], 0.0 writes/s [6066 writes total]
service time 0.000ms/operation (of which read 0.000ms, write 0.000ms)
data collected Mon, 14 Jul 2025 16:28:31

Process 'bash_script'
status OK
monitoring status Monitored

trinpc192008@localhost:~ 1 / 4
ie the mouse pointer inside or press Ctrl+G.
```

iii. Sample Configuration – Monitoring Directory

Step 1: Use the following command to create a new Monit configuration file

Example: *monitor /var/log/nginx directory for changes*

```
sudo nano /etc/monit.d/nginx_directories
```

#Add the following configuration

```
check directory nginx_log_dir with path /var/log/nginx
```

```
if failed permission 755 then alert
```

```
if failed uid 0 then alert
```

```
if failed gid 0 then alert
```

#Directory has correct permissions (755)

#Owner is user uid 0 (ie root)

#Owner group is gid 0 (ie root)

```
Applications Places Terminal
trinpce192008@localhost:~
File Edit View Search Terminal Help
GNU nano 2.3.1 File: /etc/monit.d/nginx_directories Modified
check directory nginx_log_dir with path /var/log/nginx
  if failed permission 755 then alert
  if failed uid 0 then alert
  if failed gid 0 then alert

^G Get Help    ^O WriteOut    ^R Read File   ^Y Prev Page   ^K Cut Text    ^C Cur Pos
^X Exit        ^J Justify     ^W Where Is    ^V Next Page   ^U UnCut Text  ^T To Spell
trinpce192008@localhost:~ Monit: localhost.localdomain - M... 1 / 4
```

Step 2: Reload Monit and verify the status

#Reload monit for new configuration.

sudo monit -t

sudo monit reload

sudo monit status

#Check status monit.

Monit is monitoring the **nginx_log_dir directory (i.e. **/var/log/nginx**) and found 3 errors:*

GID failed: *The directory does not belong to the root group (current GID is 990 instead of 0).*

Permission failed: *The access permission is 700, while the configuration requires 755.*

UID failed: *The directory is not owned by the root user (current UID is 992 instead of 0).*

This error occurs because Monit configuration is strictly checking the permissions and ownership of the directory.

```
Applications  Places  Terminal
trinpc192008@localhost:~
File Edit View Search Terminal Help
[trinpce192008@localhost ~]$ sudo nano /etc/monit.d/nginx_directories
[trinpce192008@localhost ~]$ sudo monit -t
Control file syntax OK
[trinpce192008@localhost ~]$ sudo monit reload
Reinitializing monit daemon
[trinpce192008@localhost ~]$ sudo monit status
Monit 5.25.1 uptime: 12m

Program 'ssh_config_checker'
  status      Initializing
  monitoring status  Initializing
  monitoring mode  active
  on reboot      start
  data collected  Tue, 15 Jul 2025 00:29:45

Directory 'nginx_log_dir'
  status      GID failed | Permission failed | UID failed
  monitoring status  Monitored
  monitoring mode  active
  on reboot      start
  permission      700
  uid             992
  gid             990
  access timestamp  Mon, 14 Jul 2025 16:17:01
  change timestamp  Sat, 05 Jul 2025 19:23:28
```

3. Program and Script Monitoring

a. Syntax Specification

CHECK PROGRAM <unique name> **PATH** <executable file> [**TIMEOUT** <number> **SECONDS**]

Description of Parameters:

CHECK PROGRAM: Monitor the result of running a script or command.

<unique name>: Unique identifier for this check.

PATH: Full path to the executable script or program.

<executable file>: The full path to the executable script or program that Monit will run periodically.

TIMEOUT (optional): Maximum allowed execution time (in seconds). If exceeded, Monit treats it as a failure.

b. Example Configuration

Step 1: Create a custom script to monitor disk usage

```
sudo nano /usr/local/bin/check_disk.sh #Create a shell script
```

#Insert the following content

```
#!/bin/bash
USED=$(df / | grep / | awk '{ print $5 }' | sed 's/%//g')
if [ $USED -gt 90 ]; then echo "Disk usage high: $USED%"
    exit 1
fi
exit 0
```

Step 2: Grant execution permission

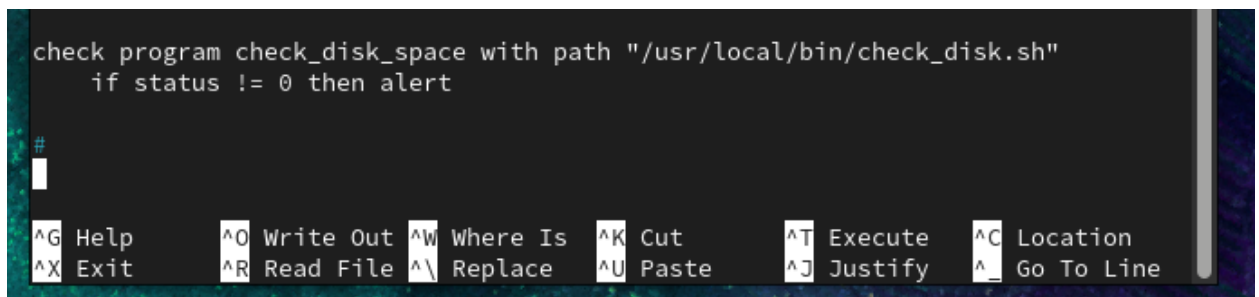
```
sudo chmod +x /usr/local/bin/check_disk.sh
```

Step 3: Configure Monit to execute the script

```
sudo nano /etc/monitrc #Edit the Monit configuration.
```

#Add the following line

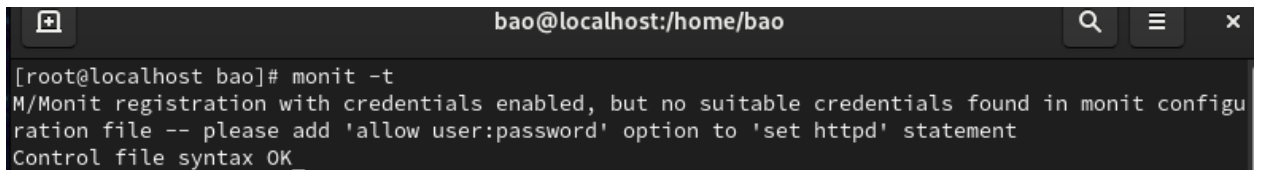
```
check program check_disk_space with path "/usr/local/bin/check_disk.sh"
if status != 0 then alert
w
```



Monit will run this script periodically. If disk usage exceeds 90%, the script exits with status 1 and Monit will log an alert.

Step 4:

systemctl monit -t *#Check for syntax errors.*

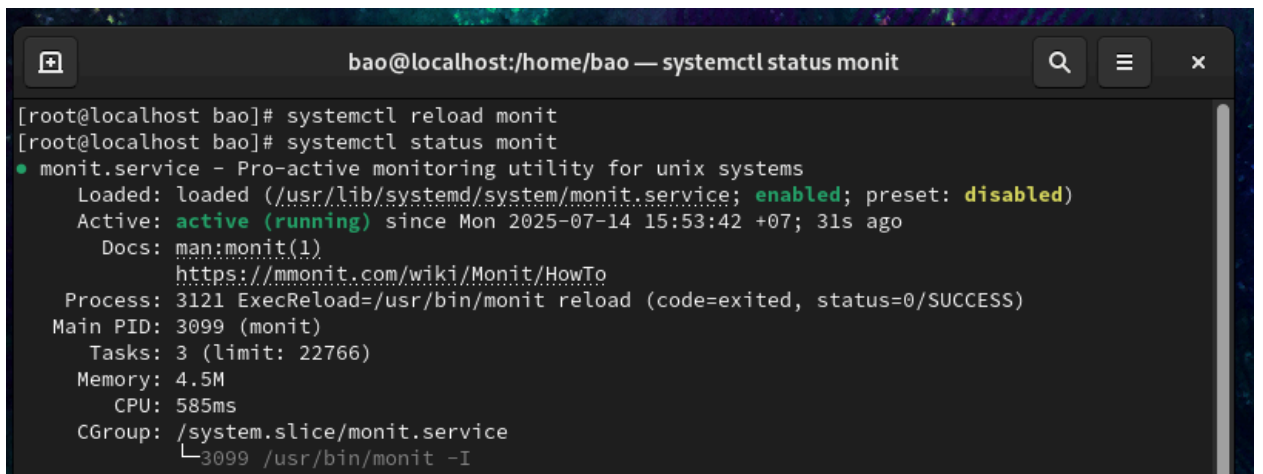
A terminal window titled 'bao@localhost:/home/bao' showing the command '[root@localhost bao]# monit -t'. The output is: 'M/Monit registration with credentials enabled, but no suitable credentials found in monit configuration file -- please add 'allow user:password' option to 'set httpd' statement' followed by 'Control file syntax OK' on a new line.

```
[root@localhost bao]# monit -t
M/Monit registration with credentials enabled, but no suitable credentials found in monit configuration file -- please add 'allow user:password' option to 'set httpd' statement
Control file syntax OK
```

If no errors are found, reload Monit.

systemctl monit reload

systemctl monit status

A terminal window titled 'bao@localhost:/home/bao — systemctl status monit' showing the command '[root@localhost bao]# systemctl reload monit' followed by '[root@localhost bao]# systemctl status monit'. The output shows the status of the 'monit.service' as 'active (running)' with various details like loaded path, active time, docs, process ID, main PID, tasks, memory, CPU, and CGroup.

```
[root@localhost bao]# systemctl reload monit
[root@localhost bao]# systemctl status monit
● monit.service - Pro-active monitoring utility for unix systems
   Loaded: loaded (/usr/lib/systemd/system/monit.service; enabled; preset: disabled)
   Active: active (running) since Mon 2025-07-14 15:53:42 +07; 31s ago
     Docs: man:monit(1)
           https://mmonit.com/wiki/Monit/HowTo
   Process: 3121 ExecReload=/usr/bin/monit reload (code=exited, status=0/SUCCESS)
    Main PID: 3099 (monit)
       Tasks: 3 (limit: 22766)
      Memory: 4.5M
         CPU: 585ms
       CGroup: /system.slice/monit.service
              └─3099 /usr/bin/monit -I
```

This configuration enables monitoring without requiring email alerts. Alerts will be recorded in the Monit log file, typically located at **/var/log/monit.log**.

```

bao@localhost:/home/bao — systemctl status monit
monit.service - Pro-active monitoring utility for unix systems
  Loaded: loaded (/usr/lib/systemd/system/monit.service; enabled; preset: disabled)
  Active: active (running) since Tue 2025-07-08 10:53:38 +07; 21s ago
    Docs: man:monit(1)
          https://mmonit.com/wiki/Monit/HowTo
  Main PID: 3699 (monit)
    Tasks: 2 (limit: 22766)
  Memory: 4.4M
    CPU: 38ms
  CGroup: /system.slice/monit.service
          └─3699 /usr/bin/monit -I

Jul 08 10:53:38 localhost.localdomain systemd[1]: Started Pro-active monitoring utility for unix system>
Jul 08 10:53:38 localhost.localdomain monit[3699]: Starting Monit 5.33.0 daemon with http interface at >
Jul 08 10:53:38 localhost.localdomain monit[3699]: 'localhost.localdomain' Monit 5.33.0 started
Jul 08 10:53:38 localhost.localdomain monit[3699]: 'nginx' process is not running'
Jul 08 10:53:38 localhost.localdomain monit[3699]: 'nginx' trying to restart
Jul 08 10:53:38 localhost.localdomain monit[3699]: 'nginx' start: '/usr/bin/systemctl start nginx'

lines 1-18/18 (END)

```

4. System Resource and Service Monitoring

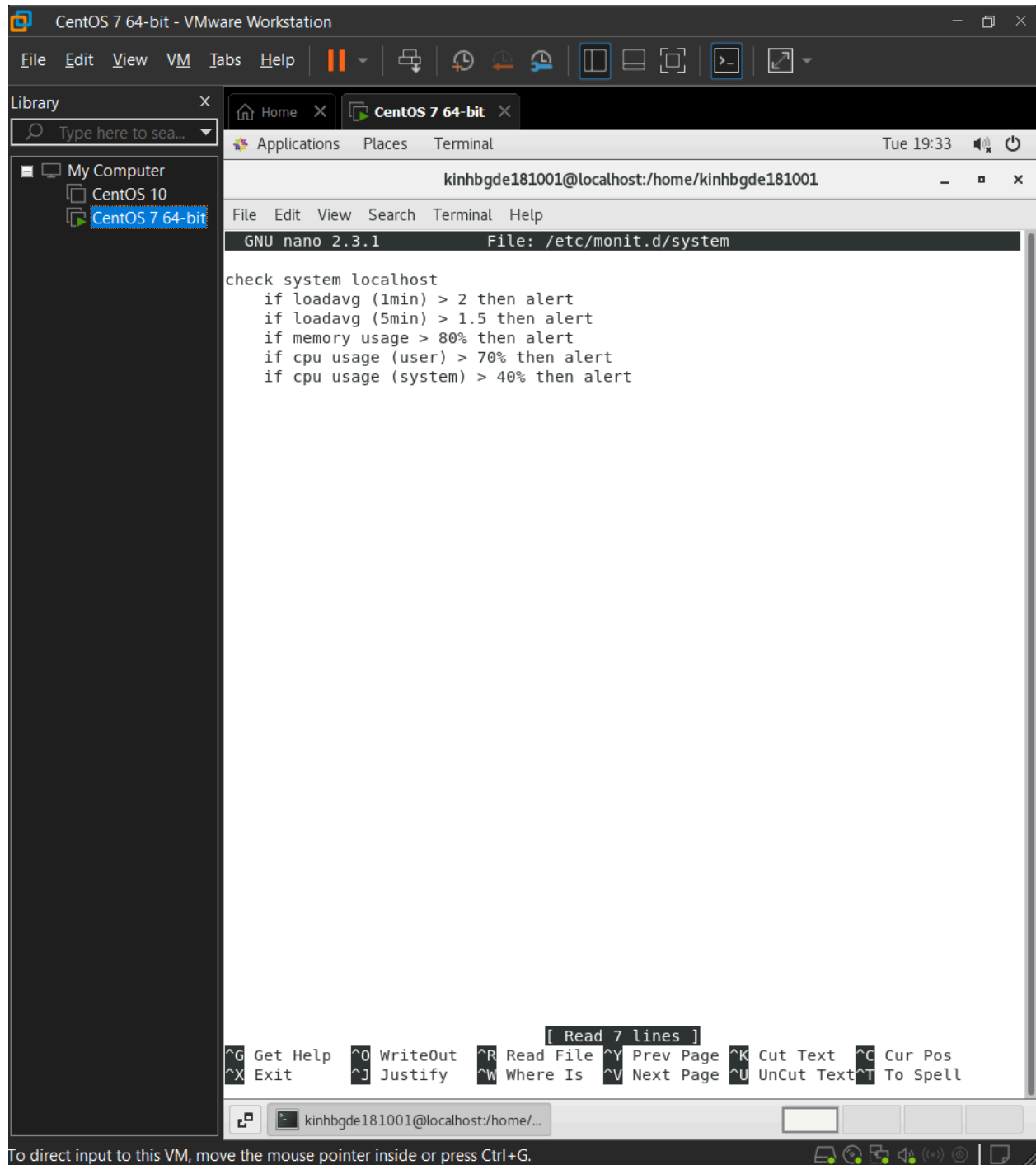
Individual monitoring configurations are typically placed in separate files within the */etc/monit.d/* directory.

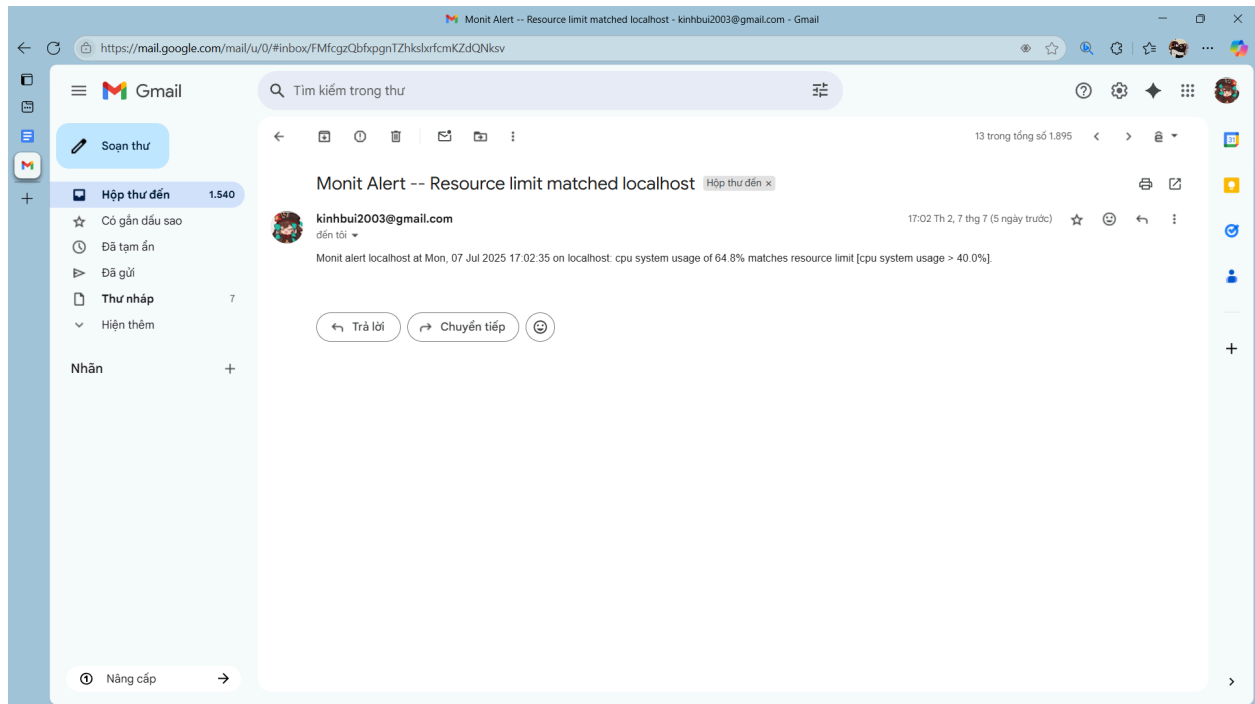
a. CPU, Memory, and Load Average Monitoring

Step 1: Create file: */etc/monit.d/system*

Step 2: Add the following lines to the file

check system localhost	<i>#Monitors overall system resources.</i>
if loadavg (1min) > 2 then alert	<i>#Alerts if 1-minute load average exceeds 2.</i>
if loadavg (5min) > 1.5 then alert	<i>#Alerts if 5-minute load average exceeds 1.5</i>
if memory usage > 80% then alert	<i>#Alerts if memory usage goes above 80%.</i>
if cpu usage (user) > 70% then alert	<i>#Alerts if user CPU usage exceeds 70%.</i>
if cpu usage (system) > 40% then alert	<i>#Alerts if system CPU usage exceeds 40%.</i>





b. Network Interface Monitoring

Step 1: Create file: **/etc/monit.d/network**

Step 2: Add the following lines to the file

check network net0 with interface ens33 *#Monitors the network interface*

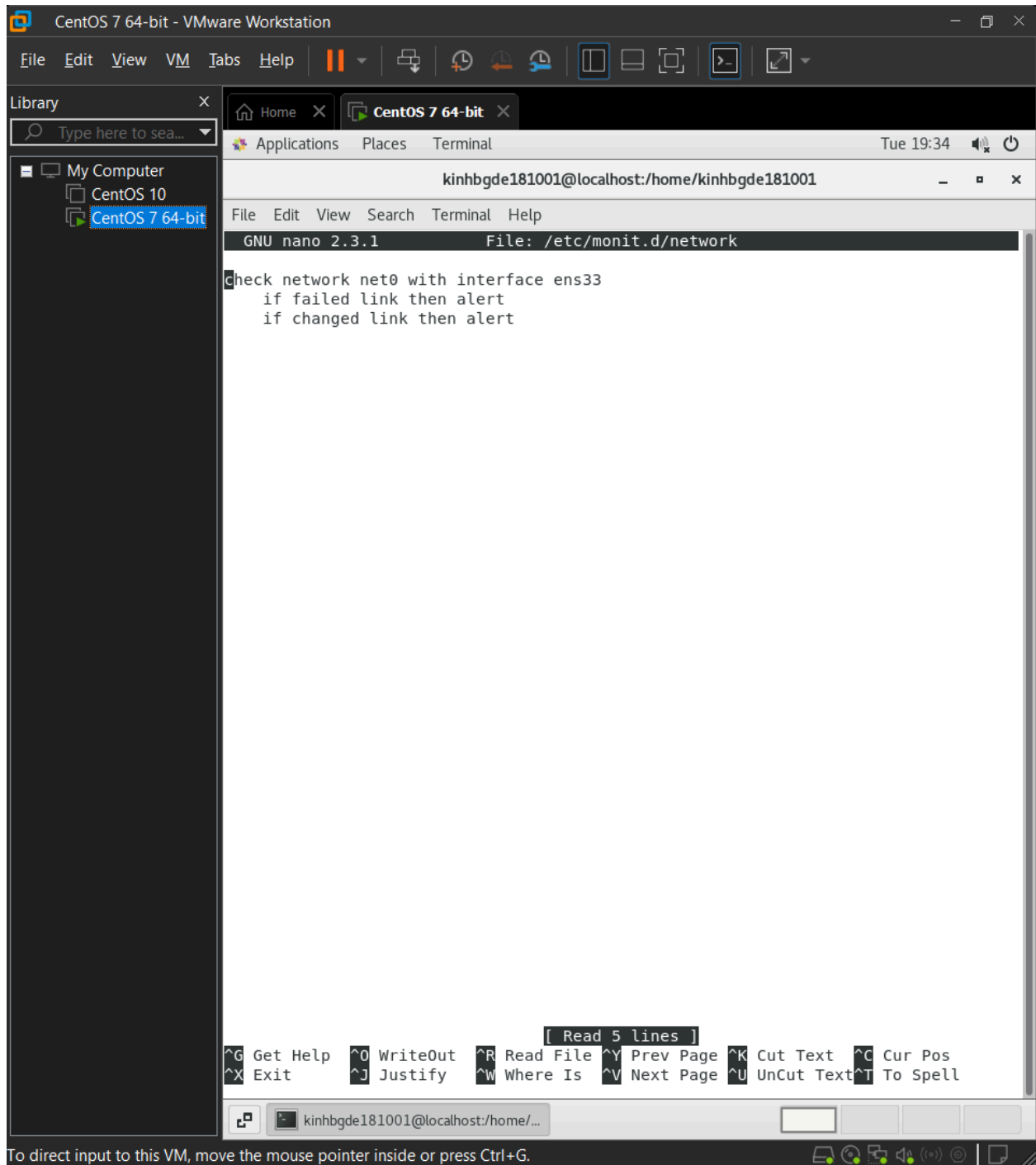
*(replace **ens33** with your actual interface name, found using **ip a**).*

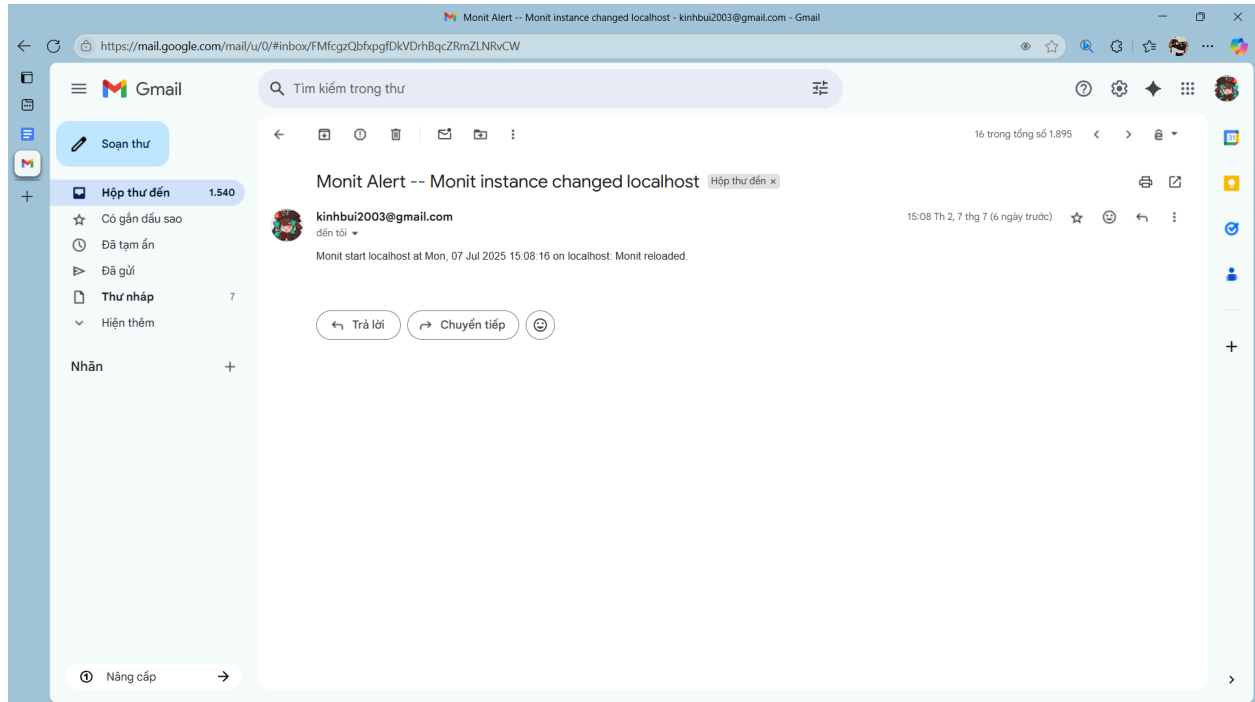
if failed link then alert

#Alerts if the network link fails.

if changed link then alert

#Alerts if the network link status changes.





c. Sendmail Service Monitoring

Step 1: Create file: **/etc/monit.d/sendmail**

Step 2: Add the following lines to the file

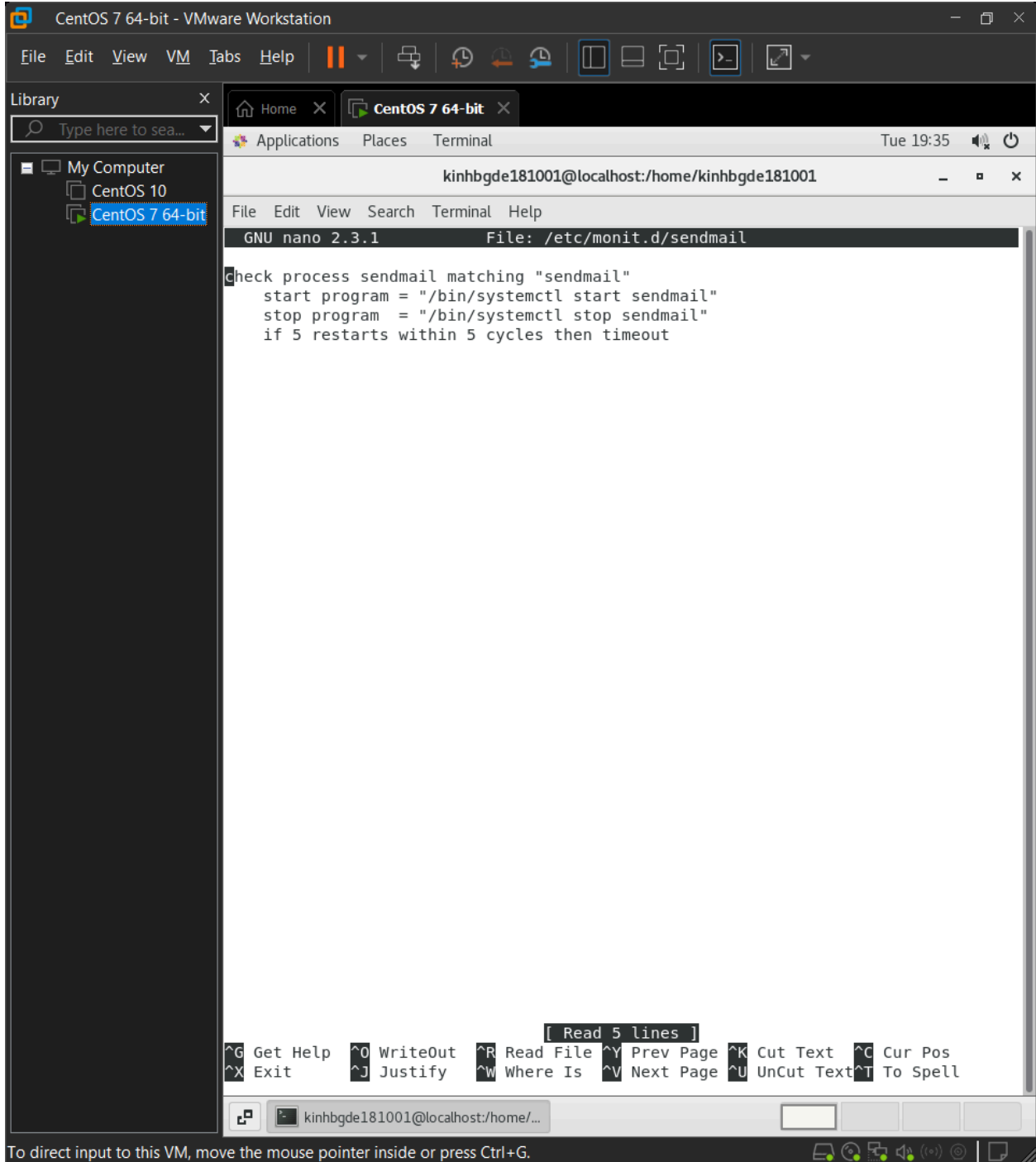
check process sendmail matching "sendmail" *#Monitors the **Sendmail** process.*

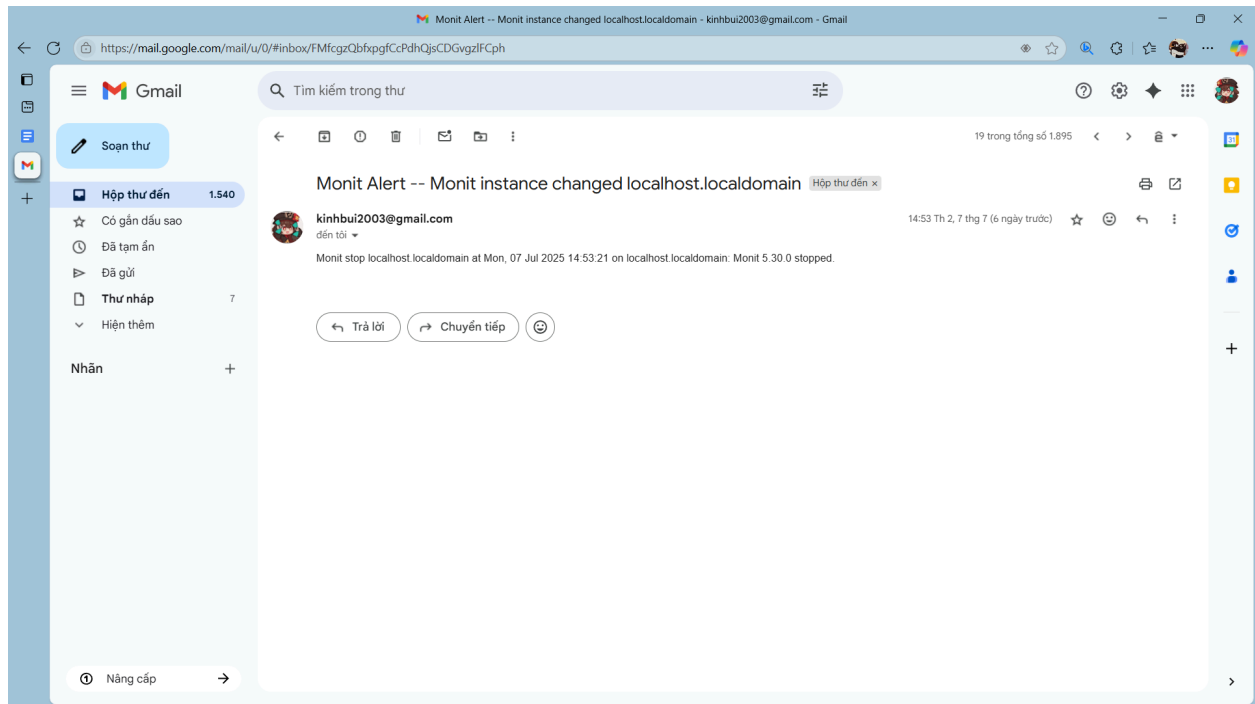
start program = "/bin/systemctl start sendmail" *#Command to **start Sendmail**.*

stop program = "/bin/systemctl stop sendmail" *#Command to **stop Sendmail**.*

if 5 restarts within 5 cycles then timeout

#If Sendmail restarts 5 times in 5 cycles, it's marked as "timeout" and Monit stops trying to restart it.





d. Apache Service Monitoring

Step 1: Create file: **/etc/monit.d/apache**

Step 2: Add the following lines to the file

check process apache with pidfile /var/run/httpd/httpd.pid

*#Monitors the **Apache process** using its **PID file**.*

start program = "/bin/systemctl start httpd" *#Command to **start Apache**.*

stop program = "/bin/systemctl stop httpd" *#Command to **stop Apache**.*

if cpu > 60% for 2 cycles then restart

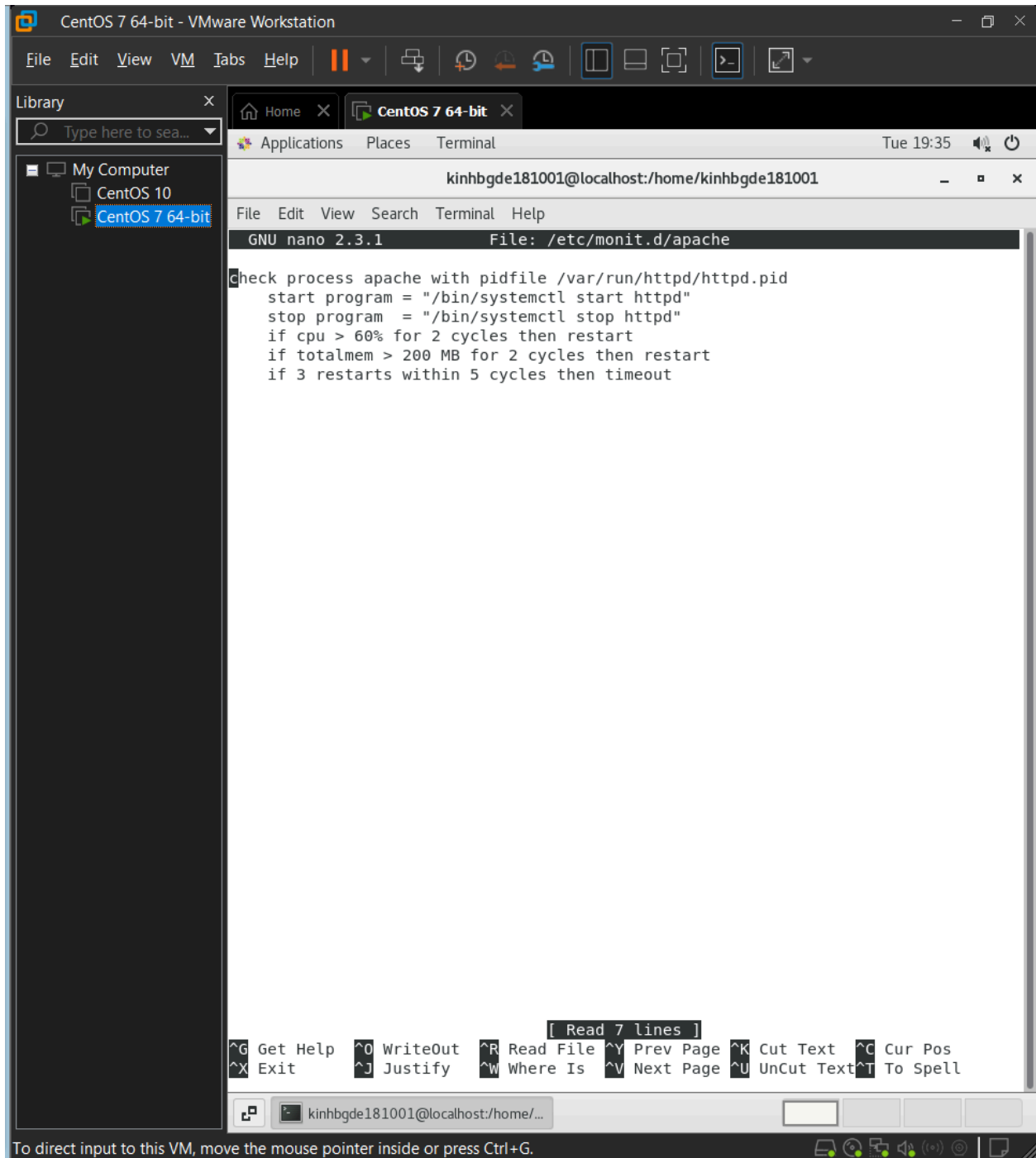
*#Restarts Apache if **CPU usage** exceeds 60% for two cycles.*

if totalmem > 200 MB for 2 cycles then restart

*#Restarts Apache if **total memory usage** exceeds 200 MB for two cycles.*

if 3 restarts within 5 cycles then timeout

#If Apache restarts 3 times in 5 cycles, it's marked as "timeout."

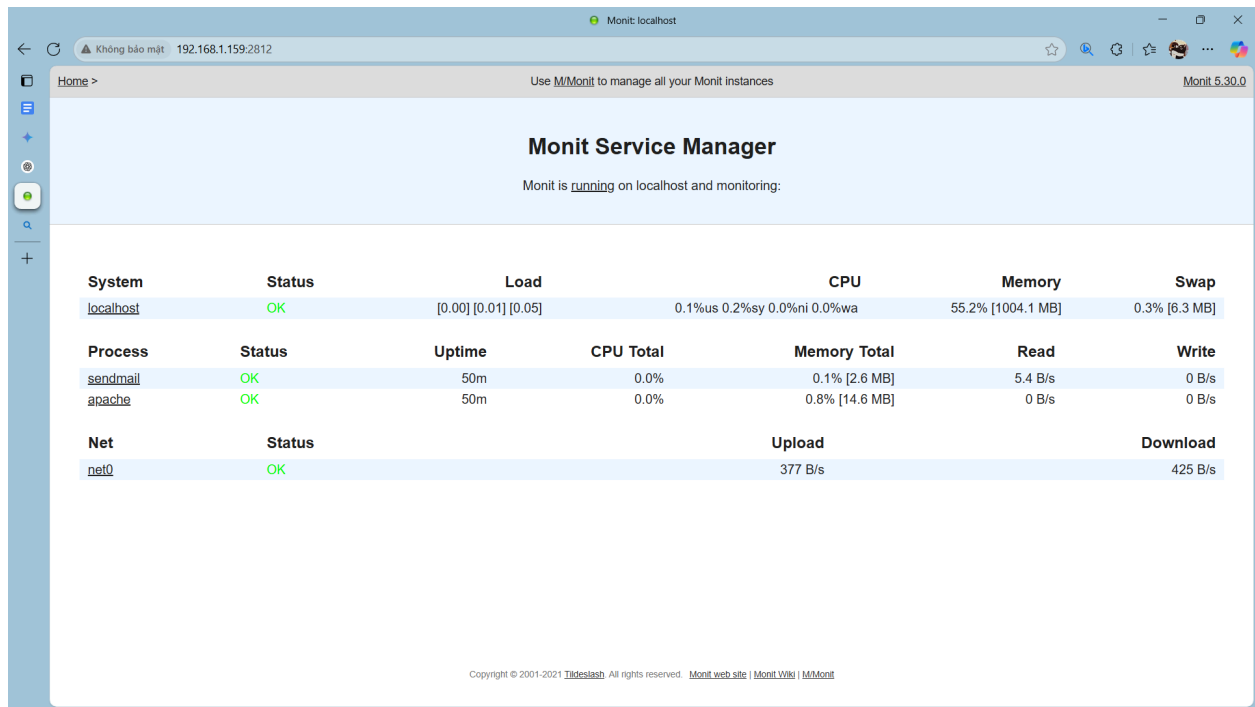


Step 3: Access the interface from the real machine to view

The virtual machine is configured in Bridged network mode.

When the IP address is 192.168.1.159, it can be accessed from the real machine by:

<http://192.168.1.159:2812>



IV. Extension: Centralized Monitoring with M/Monit

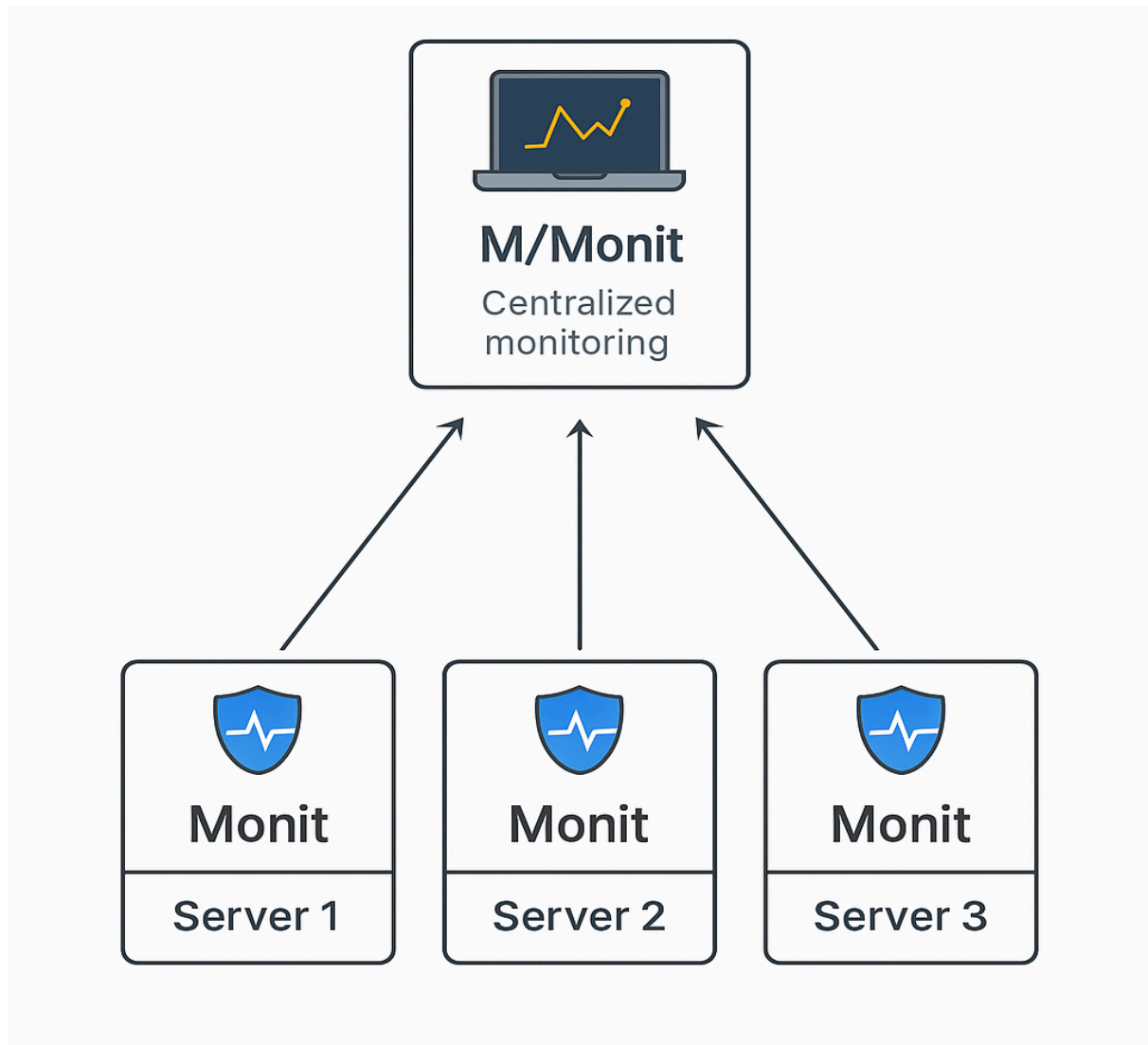
Monit is designed to monitor individual servers, so when managing multiple systems, monitoring becomes difficult.

M/Monit is a tool with a professional interface, allowing administrators to view all servers running Monit in one place for easier monitoring and management.

This section describes how to install M/Monit and link it to Monit for centralized monitoring.

***Note:** *M/Monit is a commercial tool with a 30-day free trial and is recommended to run on CentOS 8 or newer.*

Operational diagram:



Configuration on the M/Monit Server

Step 1: Download M/Monit

Visit the official M/Monit website and download the appropriate version. For example, to download version 4.3.4 for 64-bit Linux:

wget https://mmonit.com/dist/mmonit-4.3.4-linux-x64.tar.gz

```
[root@localhost opt]# wget https://mmonit.com/dist/mmonit-4.3.4-linux-x64.tar.gz
--2025-07-12 18:44:35-- https://mmonit.com/dist/mmonit-4.3.4-linux-x64.tar.gz
Resolving mmonit.com (mmonit.com)... 88.99.240.67
Connecting to mmonit.com (mmonit.com)[88.99.240.67]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 12452004 (12M) [application/x-gzip]
Saving to: 'mmonit-4.3.4-linux-x64.tar.gz'

mmonit-4.3.4-linux- 100%[=====>] 11.88M  6.61MB/s   in 1.8s

2025-07-12 18:44:40 (6.61 MB/s) - 'mmonit-4.3.4-linux-x64.tar.gz' saved [12452004/12452004]

[root@localhost opt]#
```

```
[root@localhost opt]# tar -xzf mmonit-4.3.4-linux-x64.tar.gz
[root@localhost opt]#
```

Step 2: Extract the Archive

Unpack the downloaded file using:

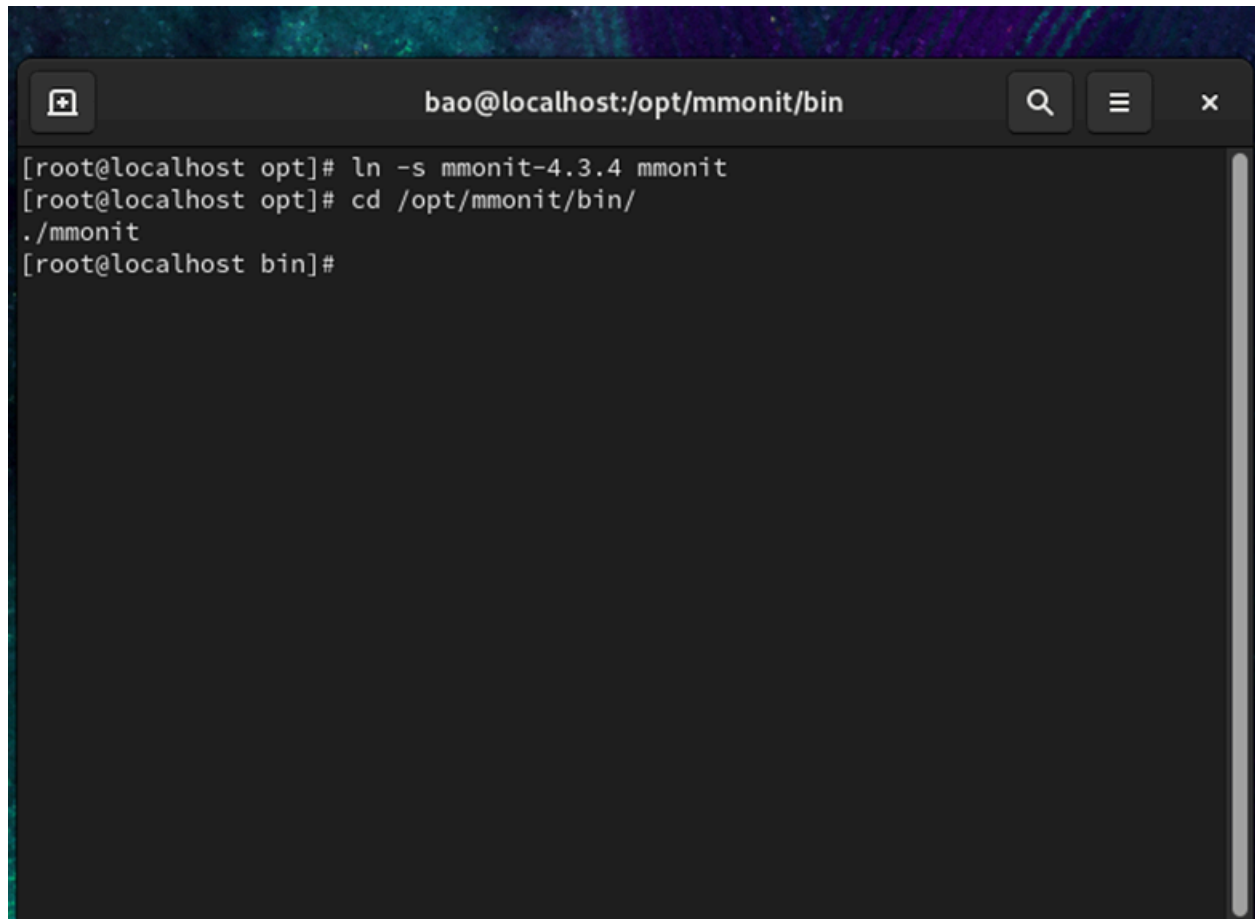
tar -xzf mmonit-4.3.4-linux-x64.tar.gz

```
bao@localhost:/opt
root@localhost opt]# ln -s mmonit-4.3.4 mmonit
root@localhost opt]#
```

Step 3: Create a Symbolic Link (Optional)

For easier directory access:

ln -s mmonit mmonit-4.3.4-linux-x64 mmonit

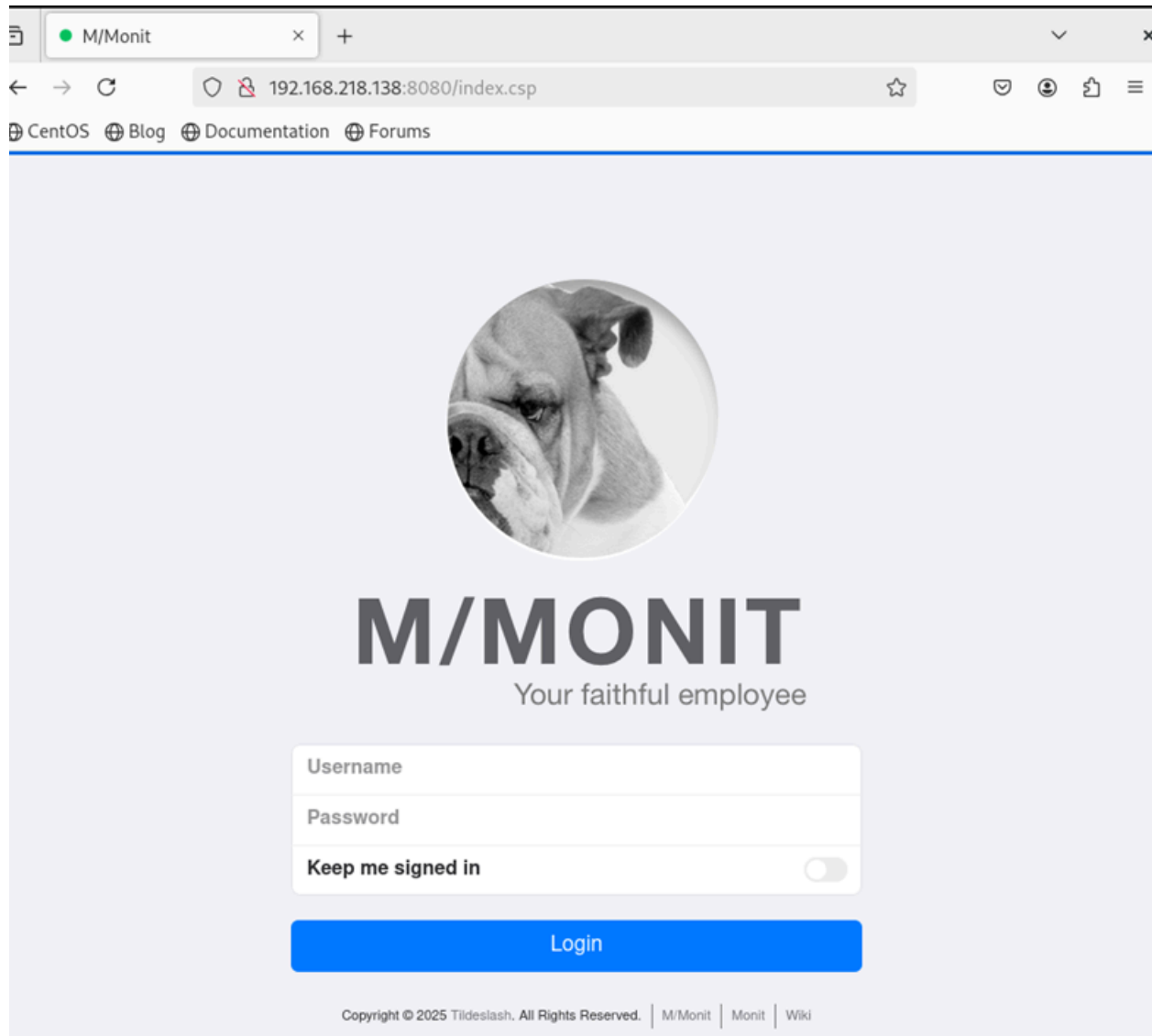
A terminal window with a dark background and a colorful, abstract pattern in the top-left corner. The window title bar shows 'bao@localhost:/opt/mmonit/bin' and standard window controls (search, menu, close). The terminal content shows the following commands and output:

```
[root@localhost opt]# ln -s mmonit-4.3.4 mmonit
[root@localhost opt]# cd /opt/mmonit/bin/
./mmonit
[root@localhost bin]#
```

Step 4: Open Firewall Port 8080

M/Monit uses port 8080 by default. Allow access to this port:

```
firewall-cmd --permanent --add-port=8080/tcp
firewall-cmd --reload
```



Configuration on Monit Agents

Step 5: Configure stations/agents running Monit to send data to M/Monit.

Edit the Monit configuration file:

```
sudo nano /etc/monitrc
```

Add the following line, replacing placeholders with actual values:

```
set mmonit http://admin:swordfish@M-MONIT-SERVER-IP:8080/collector
```

Reload Monit to update the new configuration.

```
monit -t      # Test configuration syntax.
```

```
monit reload # Reload configuration.
```

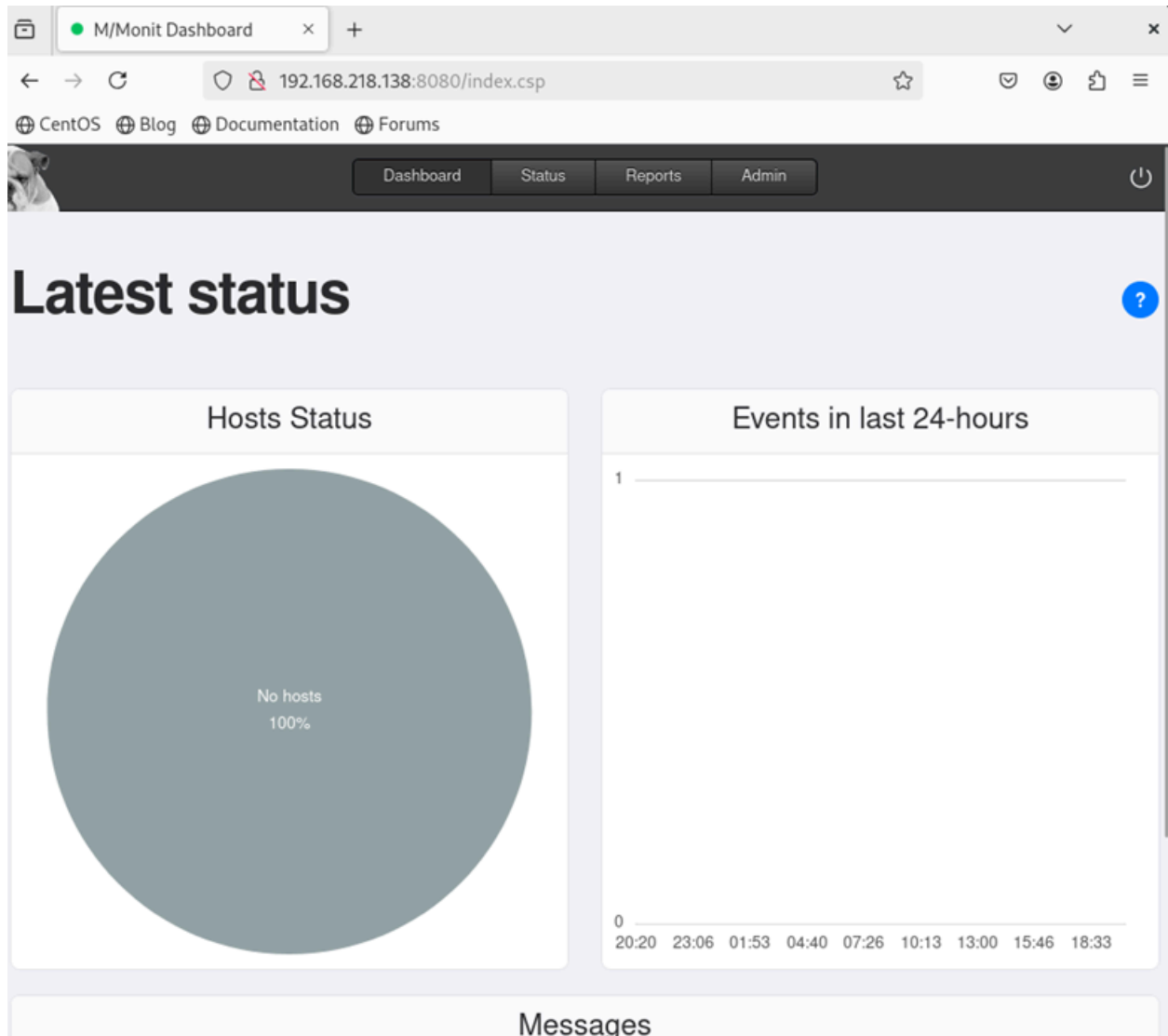
```
[root@localhost bin]# monit -t
/etc/monitrc:164: The 'address' option can be specified only once, the last value will be used
'0.0.0.0'
M/Monit registration with credentials enabled, but no suitable credentials found
in monit configuration file -- please add 'allow user:password' option to 'set
httpd' statement
Control file syntax OK
[root@localhost bin]# monit reload
/etc/monitrc:164: The 'address' option can be specified only once, the last value will be used
'0.0.0.0'
M/Monit registration with credentials enabled, but no suitable credentials found
in monit configuration file -- please add 'allow user:password' option to 'set
httpd' statement
Reinitializing monit daemon
[root@localhost bin]#
```

Step 6: Access the M/Monit Web Interface

Open a web browser and go to:

<http://<M-MONIT-SERVER-IP>:8080>

Log in using the administrator account specified earlier. The dashboard will display a centralized view of all connected Monit agents.



V. Conclusion

This report has successfully implemented an automated service monitoring system using Monit. Monit has decreased the amount of manual administration work and increased system stability.

Because of its automation and low resource consumption, Monit is a useful monitoring tool for Linux systems running on small and medium-sized servers.

However, in complicated monitoring environments or large-scale systems where distributed monitoring or sophisticated visualization capabilities are needed, this method might not be sufficient.

Feature	Description / Benefit
Lightweight and Fast	Requires minimal system resources.
Service Monitoring	Automatically restarts failed services.
Email Alert System	Sends notifications to system admins when errors occur.
Web-based Interface	Accessible interface via browser with login credentials.
File and Directory Monitoring	Detects changes in file size, timestamp, or permissions.
CPU/Memory Usage Alert	Alerts or restarts services when exceeding thresholds.
Script Execution Support	Can run custom scripts upon specific events.

VI. References

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- [3] Linux Console Vietnam, “Cài đặt Monit trên CentOS 8/Rocky 8/Alma 8/RHEL 8,” *Linux Console*, 2023. [Online]. Available: <https://vi.linux-console.net/?p=20950>. [Accessed: Jul. 11, 2025].